MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.

Each subject is divided into 2 sections:

Section 1: Single Correct Type Questions (4 options, only one correct)

Section 2: Numerical Value Type Questions (answer to be entered as numerical value)

Marking Scheme:

Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.

Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

The number of turns in primary and secondary coils of a transformer is 50 and 200 respectively. If the current in the primary coil is $4\ \mathrm{A}$, then the current in the secondary coil is

(A) $1\ \mathrm{A}$

(B) $2\ \mathrm{A}$

(C) $4\ \mathrm{A}$

(D) $5\ \mathrm{A}$

Q.2

The position of the direct image obtained at O , , when a monochromatic beam of light is passed through a plane transmission grating at normal incidence is shown in figure. The diffracted images A, B and C correspond to the first, second and third order diffraction. When the source is replaced by another source of shorter wavelength

(A) all the four will shift in the direction C to O

(B) all the four will shift in the direction O to C

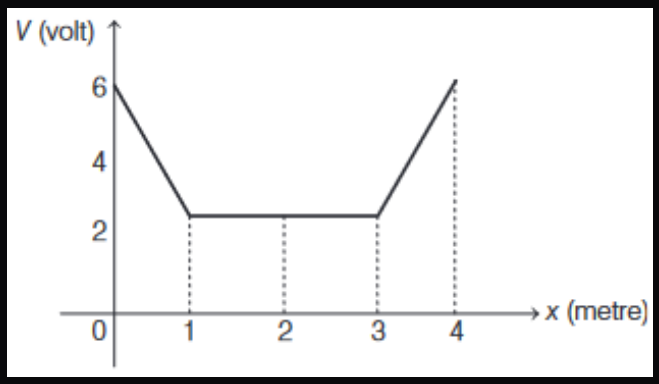
(C) the images C,B and A will shift towards O

(D) the images C,B and A will shift away from O

Q.3

The variation of electric potential with distance from a fixed point is shown in figure. What is the value of electric

field at $x = 2$ m ?



(A) 0

(B) 3

(C) 2

(D) 6

Q.4

On six complete rotations, the screw gauge moves by 3 mm on the main scale. If there are 50 divisions on the circular scale the least count of the screw gauge is

(A) 0.01 cm

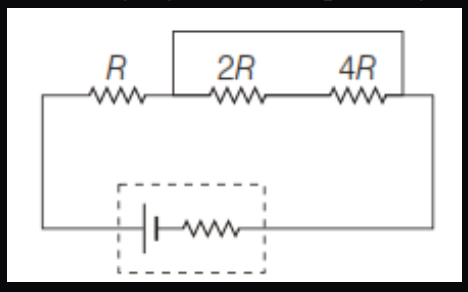
(B) 0.02 mm

(C) 0.001 mm

(D) 0.001 cm

Q.5

A battery with internal resistance of 4Ω is connected to a circuit consisting three resistances, $R, 2R$ and $4R$ (see following figure). If the power generated in the circuit is highest, then the magnitude of R must be



(A) 4Ω

(B) 7Ω

(C) 10Ω

(D) 14Ω

Q.6

When an inductor L and a resistor R in series are connected across a $12\ \mathrm{V}, 50\ \mathrm{Hz}$ supply, a current of 0.5 A flows in the circuit. The current differs in the phase from applied voltage by $\frac{\pi}{3}$ radian. Then the value of R is

(A) 10Ω

(B) 3Ω

(C) 12Ω

(D) 15Ω

Q.7

A body is projected at an angle θ so that its range is maximum. If T is the time of flight, then the value of maximum range is (acceleration due to gravity $= g$)

(A) $\frac{g^2T}{2}$

(B) $\frac{gT}{2}$

(C) $\frac{gT^2}{2}$

(D) $\frac{g^2T^2}{2}$

Q.8

The equation that represents magnetic field of a plane electromagnetic wave which is propagating along x-direction with wavelength $10\ \mathrm{mm}$ and maximum electric field $60\mathrm{Vm}^{-1}$ in y-direction is (where, $c =$ speed of light)

(A) $\left(6 \times 10^{-7}\right) \sin[0.2\pi(ct - x)]\hat{k}T$

(B) $\left(2 \times 10^{-7}\right) \sin[200\pi(ct - x)]\hat{k}T$

(C) $\left(2 \times 10^{-7}\right) \sin[200\pi(ct - x)]\hat{i}\ \mathrm{T}$

(D) $\left(6 \times 10^{-7}\right) \sin[02\pi(ct - x)]\hat{i}T$

Q.9

A photon of wavelength 300 nm interacts with a stationary hydrogen atom in ground state. During the interaction, whole energy the photon is transferred to the electron of the atom. State which possibility is correst. (Consider, Plank constant $= 4 \times 10^{-15} eV_s$, velocity of light $= 3 \times 10^8 \mathrm{m/s}$, ionisation energy of hydrogen $= 13.6\mathrm{eV}$)

(A) Electron will be knocked out of the atom

(B) Electron will go to any excited state of the atom

(C) Electron will go only to lirst excited state of the atom

(D) Electron will keep orbiting in the ground state of the atom

Q.10

When a wave traverses a medium, the displacement of a particle located at x at a time t is given by $y = a\sin(bt - cx)$, where a, b and c are constants of the wave, which of the following is a quantity with dimensions?

(A) $\frac{y}{a}$

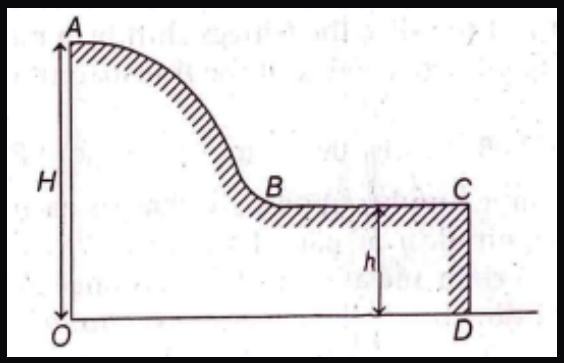
(B) bt

(C) cx

(D) $\frac{b}{c}$

Q.11

A small disc of mass m slides down with initial velocity zero from the top (A) of a smooth hill of height H having a horizontal portion (BC) as shown in the figure. If the height of the horizontal portion of the hill is h, then the maximum horizontal distance covered by the disc from the point D is



(A) $\frac{H}{2}$

(B) $2\mathrm{H}$

(C) H

(D) $3H$

Q.12

A long solenoid having 100 turns per cm carries a current of $\frac{4}{\pi}$ A. At the centre of it is placed a coil of 200 turns of cross-sectional area $25\ \mathrm{cm}^2$ having its axis parallel to the field produced by the solenoid. When the direction of the current in the solenoid is reversed with in $0.04\ \mathrm{s}$, the induced emf in the coil is

(A) $0.2\ \mathrm{V}$

(B) $0.4\ \mathrm{V}$

(C) $0.002\ \mathrm{V}$

(D) $0.016\ \mathrm{V}$

Q.13

Statement (A) As one considers orbits with higher values of n in hydrogen atom, the electric potential energy of the atom increases.
Statement (B) In thomson's model, an atom is a spherical cloud of positive charges with electrons embedded in it.

Statement (C) The orbital picture is Bohr's model of the hydrogen atom was consistent with the uncertainty principle.

(A) A, B and C are true

(B) A, B true, but C false

(C) B, C true, but A false

(D) A, C true but B false

Q.14

The temperature of 5 moles of a gas at constant volume is changed from $100^{\circ}\mathrm{C}$ to $120^{\circ}\mathrm{C}$. The change in internal energy is $80\ \mathrm{J}$. The total heat capacity of the gas at constant volume will be in $\mathrm{J/K}$ is :

(A) 8

(B) 4

(C) 0.8

(D) 0.4

Q.15

A moving coil galvanometer of resistance 100 Ω shows full scale deflection when a current of 100 micro-amperes passes through it. If it is intended to show full scale deflection when a current of 1 milliampere passes through it, the value of shunt resistance in ohms to be connected to the galvanometer is.

(A) $\frac{9}{4}$

(B) $\frac{10}{3}$

(C) $\frac{100}{9}$

(D) $\frac{900}{7}$

Q.16

A block of mass m is placed on a smooth wedge of inclination θ. The whole system is accelerated horizontally so that the block does not slip on the wedge. The force exerted by the wedge on the block (g is acceleration due to gravity) will be

(A) $mg\cos\theta$

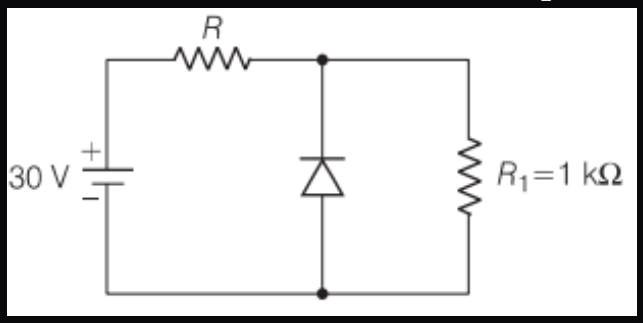
(B) $mg\sin\theta$

(C) mg

(D) $mg\sec\theta$

Q.17

If current in diode is five times that in R_1 and breakdown voltage of diode is $6\ \text{V}$, find the value of R ?



(A) 2000Ω

(B) $\frac{2000}{3}\Omega$

(C) 1000Ω

(D) $\frac{1000}{3}\Omega$

Q.18

A particle of mass M is moving in a horizontal circle of radius R with uniform speed V. When the particle moves from one point to a diametrically opposite point, its

(A) momentum does not change

(B) momentum changes by $2MV$

(C) kinetic energy changes by $\frac{\text{MV}^2}{4}$

(D) kinetic energy changes by MV^2

Q.19

The rms speed of oxygen molecule at a certain temperature is $600\ \text{ms}^{-1}$. If the temperature is doubled and oxygen molecule dissociates into atomic oxygen atoms, the new rms speed is

(A) $120\ \text{ms}^{-1}$

(B) $150\ \text{ms}^{-1}$

(C) $1200\ \text{ms}^{-1}$

(D) $600\ \text{ms}^{-1}$

Q.20

Two bodies of masses m_1 and m_2 initially at rest at infinite distance apart move towards each other under gravitational force of attraction. Their relative velocity of approach when they are separated by a distance r is (G = universal gravitational constant.)

(A) $\left[\frac{2G(m_1-m_2)}{r}\right]^{1/2}$

(B) $\left[\frac{2G(m_1+m_2)}{r}\right]^{1/2}$

(C) $\left[\frac{r}{2G(m_1m_2)}\right]^{1/2}$

(D) $\left[\frac{r}{2G}m_1m_2\right]^{1/2}$

Q.21

A $20\ \mathrm{F}$ capacitor is charged to $5\ \mathrm{V}$ and isolated. It is then connected in parallel with an uncharged $30\ \mathrm{F}$ capacitor. The decrease in the energy of the system will be

Q.22

The intensity of magnetization of a bar magnet is $5.0 \times 10^4 \mathrm{Am}^{-1}$. The magnetic length and the area of cross-section of the magnet are $12\mathrm{cm}$ and $1\mathrm{cm}^2$ respectively. The magnitude of magnetic moment of this bar magnet is (in SI unit) M, find 10M.

Q.23

In a hydrogen sample, if the atoms are excited to states with principal quantum number 20 , then the number of different wavelengths which may be observed in the spectrum is

Q.24

A car moving with uniform acceleration covers the distance of $200\ \mathrm{m}$ in first 2 seconds and the distance of $220\ \mathrm{m}$ in next 4 seconds. The velocity of the car after 7 seconds is

Q.25

A solid cylinder is released from rest from the top of an inclined plane of inclination 30° and length $60\ \mathrm{cm}$. If the cylinder rolls without slipping, then the speed when it reaches the bottom is

Q.26

Which of the following is a correct statement?

(A) Aqueous solutions of Cu^+ and Zn^{2+} are colourless

(B) Aqueous solutions of Cu^{2+} and Zn^{2+} are colourless

(C) Aqueous solution of Fe^{3+} is green in colour

(D) Aqueous solution of MnO_4^- is colourless

Q.27

Match the following.

	Column I		Column II
A.	Acetaldehyde, vinyl alcohol	1.	Enantiomers
B.	Eclipsed and staggered ethane	2.	Tautomers
C.	(+)2-butanol, (–)2-butanol	3.	Chain isomers
D.	Methyl-*n*-propyl-amine and diethylamine	4.	Conformational isomers
		5.	Metamers

A B C D

(A) $1 \quad 4 \quad 3 \quad 5$

(B) $2 \quad 4 \quad 1 \quad 5$

(C) $5 \quad 1 \quad 4 \quad 2$

(D) $5 \quad 1 \quad 3 \quad 2$

Q.28

The order of screening effect among s, p, d and f-orbitals of a given shell of an atom in its outer shell electrons is

(A) $s > p > d > f$

(B) $f > d > p > s$

(C) $p < d < s > f$

(D) $d > f > p > s$

Q.29

$[X] +$ dil. $\mathrm{H_2SO_4} \longrightarrow [Y]$:

Colourless, suffocating gas

$[Y] + \mathrm{K_2Cr_2O_7} + \mathrm{H_2SO_4} \longrightarrow$

Green colouration of solution

Then, $[X]$ and $[Y]$ are

(A) $\mathrm{SO_3^{2-}}, \mathrm{SO_2}$

(B) $\mathrm{Cl^-}, \mathrm{HCl}$

(C) $\mathrm{S^{2-}}, \mathrm{H_2S}$

(D) $\mathrm{CO_3^{2-}}, \mathrm{CO_2}$

Q.30

Assertion (A) Xe atoms in $\mathrm{XeF_2}$ are d^2sp^3 hybridised.
Reason (R) $\mathrm{XeF_2}$ molecule does not follow octet rule.
Which of the following is correct?

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A)

(C) (A) is true, but (R) is false

(D) (A) is false, but (R) is true

Q.31

Which one of the following statements is correct?

(A) Starch on complete hydrolysis gives fructose

(B) Lactose on hydrolysis gives glucose and fructose

(C) Glucose on slow oxidation to $\mathrm{CO_2}$ and $\mathrm{H_2O}$ by enzyme does not liberate energy

(D) Cellulose is not digestible in human body

Q.32

During the depression in freezing point experiment, an equilibrium is established between the molecules of

(A) liquid solvent and solid solvent

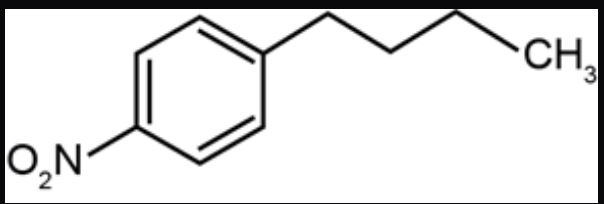
(B) liquid solute and solid solvent

(C) liquid solute and solid solute

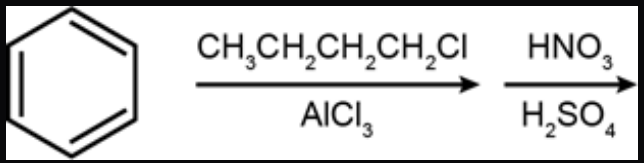
(D) liquid solvent and solid solute

Q.33

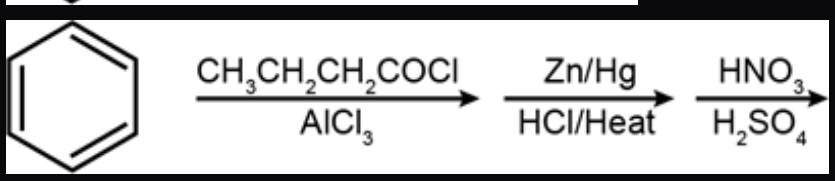
ldentify the correct method for the synthesis of lhe compound shown below from the following alternatives.



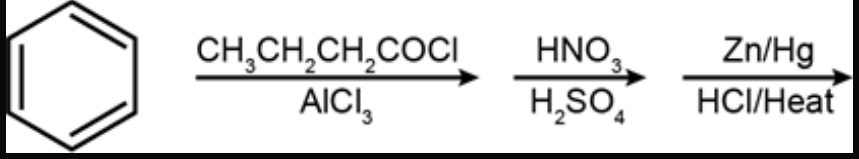
(A)



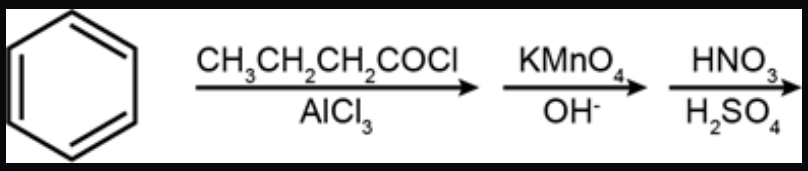
(B)



(C)



(D)



Q.34

Match the following.

	Column I		Column II
A.	sp^3	(i)	$[Co(NH_3)_6]^{3+}$
B.	dsp^2	(ii)	$[Ni(CO)_4]$
C.	sp^3d^2	(iii)	$[Pt(NH_3)_2Cl_2]$
D.	$d\ sp^3$	(iv)	$[CoF_6]^{3-}$
		(v)	$[Fe(CO)_5]$

A B C D are respectively

(A) (v) (ii) (iv) (iii)

(B) (ii) (iii) (iv) (v)

(C) (ii) (iii) (i) (v)

(D) (iii) (ii) (iv) (i)

Q.35

Which one of the following lanthanide ions does not exhibit paramagnetism?

(A) Lu^{3+}

(B) Ce^{3+}

(C) Eu^{3+}

(D) Yb^{3+}

Q.36

Assertion (A) Cyclohexane is the most stable cycloalkane.

Reason (R) Cyclopropane and cyclobutane are less stable due to angle strain and torsional strain.
The correct answer is

(A) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(B) (A) is true but (R) is not true.

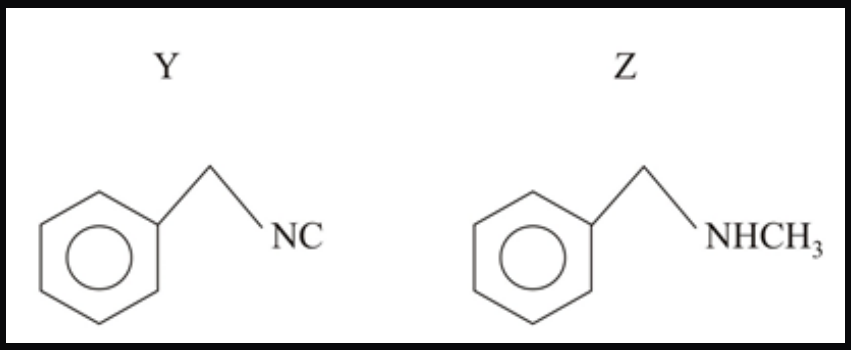
(C) (A) is not true but (R) is true.

(D) Both (A) and (R) are true and (R) is the correct explanation of (A).

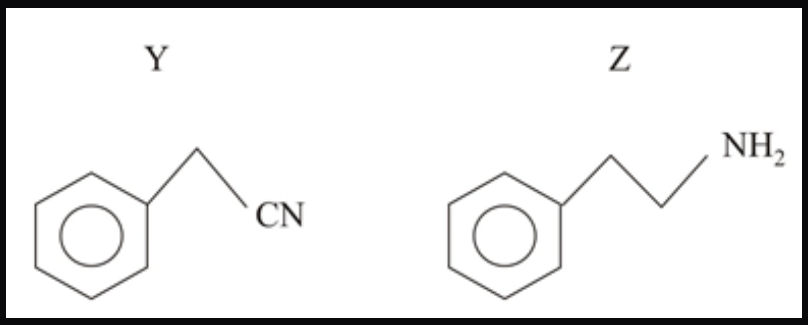
Q.37

An organic compound $X(C_7H_7Cl)$ when reacted with KCN/C_2H_5OH gave a major product Y. Z is formed when Y is reduced with $LiAlH_4$. What are Y and Z?

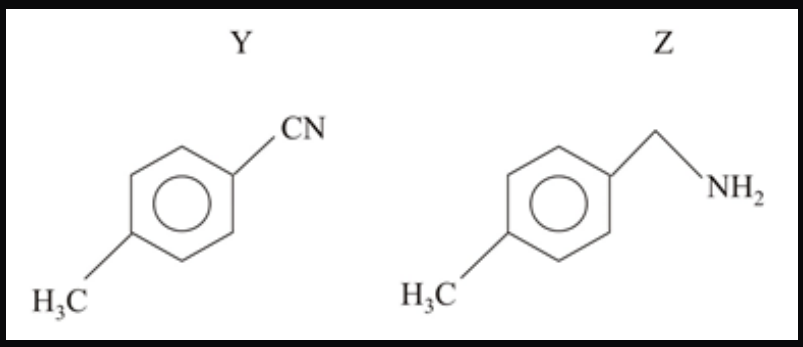
(A)



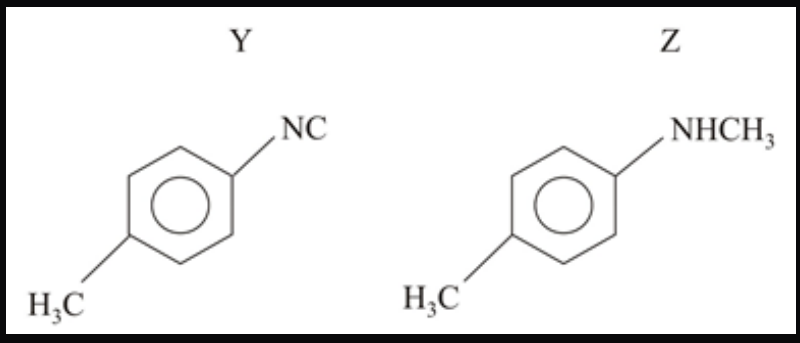
(B)



(C)

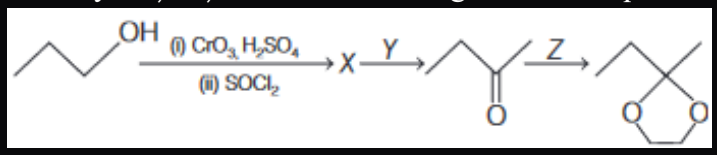


(D)

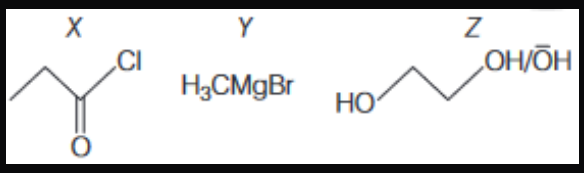


Q.38

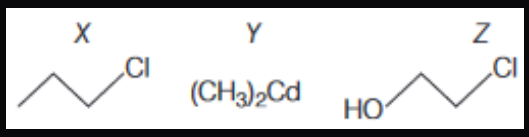
Identify X, Y, Z in the following reaction sequence



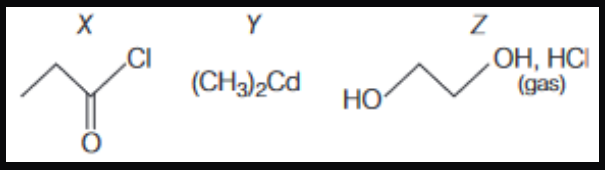
(A)



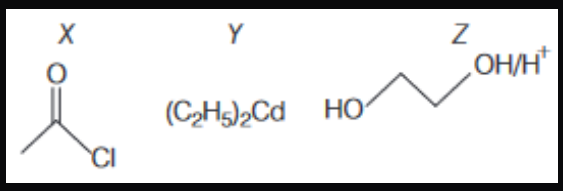
(B)



(C)



(D)



Q.39

Maximum number of electrons possible with spin quantum number $+\frac{1}{2}$ with principal quantum number $\mathrm{n} = 4$ in an atom is

(A) 16

(B) 9

(C) 4

(D) 25

Q.40

The compound which does not respond to iodoform test is

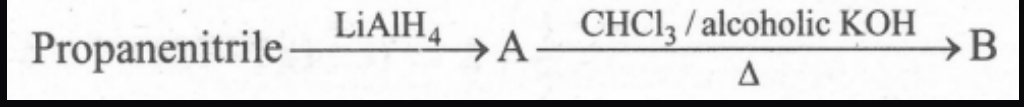
(A) $CH_3 - CHO$

(B) $CH_3CH(OH)CH_3$

(C) $C_2H_5 - CO - C_2H_5$

(D) $C_6H_5COCH_3$

Q.41



Conversion of A to B is an example of the reaction

(A) Reimer-Tiemann reaction

(B) Carbyl amine reaction

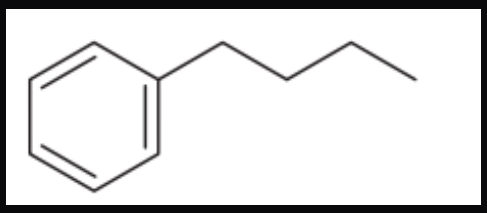
(C) Stephen reaction

(D) Sandmeyer reaction

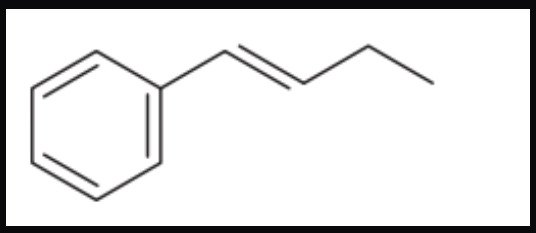
Q.42

Ethyl phenyl acetylene (1-phenyl-1-butyne) on reduction with partially deactiveated palladised charcoal (Lindlar's catalyst) gives

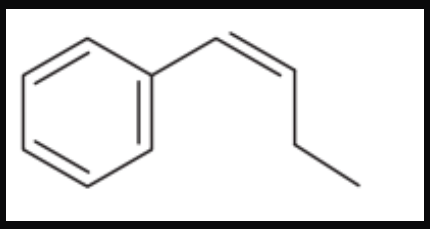
(A)



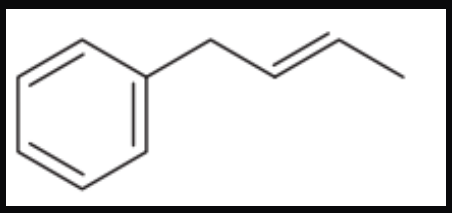
(B)



(C)

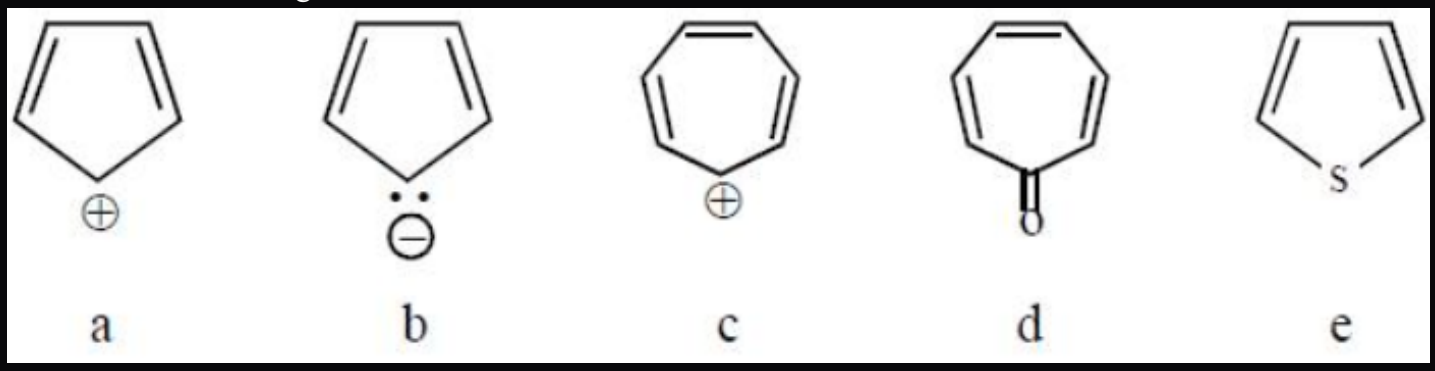


(D)



Q.43

Which of the following structures are aromatic?



(A) a, c, e

(B) b, c, d, e

(C) a, b, c

(D) b, d

Q.44

The crystal filed theory is successful in explaining which of the following?
I. Ligands as point charges;
II. Formation and structures of complexes.
III. colour;
IV. Magnetic properties:
V . covalent character of metal-ligand bonding.

(A) $\mathrm{I, II, III}$ only

(B) $\mathrm{II, III, IV}$ only

(C) $\mathrm{III, IV, V}$ only

(D) $\mathrm{II, IV, V}$ only

Q.45

Assertion (A) $[\mathrm{SiF_6}]^{2-}$ is formed but $[\mathrm{SiCl_6}]^{2-}$ is not
Reason (R) Electronegativity (EN) of F is higher than EN of Cl

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(B) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(C) (A) is correct but (R) is not correct

(D) (A) is not correct but (R) is correct

Q.46

If $5\ \mathrm{L}$ of an ideal gas at a constant external pressure of $2\ \mathrm{atm}$ expands isothermally to a final volume of $'\mathrm{X}'\ \mathrm{L}$, the system does a work of $-2,026.4\ \mathrm{J}$. $'\mathrm{X}'$ (in L) is (1 L. atm $=101.32\ \mathrm{J}$)

Q.47

Vapour density of a metal chloride is 83. If equivalent weight of the metal is 6, its atomic weight will be ____.

Q.48

According to molecular orbital theory, the total number of bonding electron pairs in O_2 is

Q.49

Consider a titration of potassium dichromate solution with acidified Mohr's salt solution using diphenyl amine as indicator. The number of moles of Mohr's salt required per mole of dichromate is

Q.50

Calculate the energy required to convert all atoms in 4.8 g of Mg to Mg^{2+} in the vapour state. IE_1 and IE_2 of Mg are $740\ kJ/mol$ and $1450\ kJ/mol$ respectively. (Mark answer in KJ/mol)

Q.51

The mean deviation from the median for the following data is

x	10	11	12	13
f	6	12	18	12

(A) 0.75

(B) 7.5

(C) 0.65

(D) 0.40

Q.52

Let g_n be the nth term of the geometric progression of positive numbers. If $\sum_{n=1}^{100} g_{2n} = \frac{10}{3}$ and $\sum_{n=1}^{100} g_{2n-1} = \frac{5}{9}$ then the common ratio of geometric progression, is

(A) 2

(B) 4

(C) 6

(D) 8

Q.53

Find number of integral values of λ if $(\lambda, \lambda+1)$ is an interior points of $\triangle ABC$, where $A \equiv (0,3), B \equiv (-2,0)$ and $C \equiv (6,1)$.

(A) 5

(B) 1

(C) 3

(D) 2

Q.54

Number of vectors $\vec{V_1} = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$ and $\vec{V_2} = a_2\hat{i} + b_2\hat{j} + c_2\hat{k}$ such that $\vec{V_1}$ and $\vec{V_2}$ are mutually perpendicular, where $\mathbf{a}_1, \mathbf{b}_1, \mathbf{c}_1, \mathbf{a}_2, \mathbf{b}_2, \mathrm{c}_2 \in \{-2,-1,1,2\}$ is equal to

(A) 288

(B) 278

(C) 164

(D) 184

Q.55

The shortest distance between the skew-lines $\mathbf{r} = (-\hat{\mathbf{i}} + 3\hat{\mathbf{k}}) + t(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}})$ and $\mathbf{r} = (3\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}) + s(2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}})$ is

(A) $\frac{10}{\sqrt{17}}$

(B) $\frac{22}{\sqrt{17}}$

(C) 9

(D) 8

Q.56

If $\mathrm{Lim}_{n\to\infty} \frac{n\cdot 3^n}{n(x-2)^n + n\cdot 3^{n+1} - 3^n} = \frac{1}{3}$ then the range of x is $(n \in N)$

(A) $[2,5)$

(B) $(1,5)$

(C) $(-1,5)$

(D) $(-\infty, \infty)$

Q.57

Suppose families always have one, two or three children, with probabilities $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{1}{4}$ respectively. Assume everyone eventually gets married and has children, the probability of a couple having exactly four grandchildren is

(A) $\frac{27}{128}$

(B) $\frac{37}{128}$

(C) $\frac{25}{128}$

(D) $\frac{20}{128}$

Q.58

Let L be the least value of the expression $\frac{\cot 2x - \tan 2x}{1+\sin\left(\frac{5\pi}{2}-8x\right)}$ if $x \in \left(0, \frac{\pi}{8}\right)$ and M denotes the value of $\frac{4\left(\cos^3 10^\circ + \sin^3 20^\circ\right)}{(\cos 10^\circ + \sin 20^\circ)}$. Find the value of $(L+M)$.

(A) 1

(B) 5

(C) 3

(D) 2

Q.59

The equation $\begin{vmatrix} (1+x)^2 & (1-x)^2 & -\left(2+x^2\right) \\ 2x+1 & 3x & 1-5x \\ x+1 & 2x & 2-3x \end{vmatrix} + \begin{vmatrix} (1+x)^2 & 2x+1 & x+1 \\ (1-x)^2 & 3x & 2x \\ 1-2x & 3x-2 & 2x-3 \end{vmatrix} = 0$

(A) Has no real solution

(B) Has 4 real solutions

(C) Has two real and two non-real solutions

(D) Has infinite number of solutions, real or non-real

Q.60

The common chord of two intersecting circles C_1 and C_2 can be seen from their centres at the angles of 90° and 60° respectively. If the distance between their centres is equal to $\sqrt{3}+1$ then the radius of C_1 and C_2 are

(A) $\sqrt{3}$ and 3

(B) $\sqrt{2}$ and $2\sqrt{2}$

(C) $\sqrt{2}$ and 2

(D) $2\sqrt{2}$ and 4

Q.61

The domain of the function
$f(x) = \sin^{-1}\left[\log_4\left(\frac{x}{4}\right)\right] + \sqrt{17x - x^2 - 16}$ is

(A) $[-1, 1]$

(B) $[1, 4]$

(C) $(0, 16]$

(D) $[1, 16]$

Q.62

The focus of an ellipse is $\mathrm{M}(-1,-1)$ and corresponding directrix $x-y+3=0$ and eccentricity is $\frac{1}{2}$. If the centre of the ellipse be $C(p, q)$, then find the value of $(2p-4q)$.

(A) 4

(B) 5

(C) 6

(D) None

Q.63

A function ' f ' is defined by $\mathrm{f}(\mathrm{z})=(4+\mathrm{i})\mathrm{z}^2+\alpha \mathrm{z}+\gamma$ for all complex number z, where α and γ are complex numbers if $f(1)$ and $f(i)$ are both real and the smallest possible values of $|\alpha|+|\gamma|$ is p then determine p^2

(A) 5

(B) 1

(C) 3

(D) 2

Q.64

Let $f(x)$ be a function with continuous derivative on the interval (λ, μ) and
$2 \ln \left(f(x)+\sqrt{(f(x))^2+1}\right) \cdot f'(x)-\sqrt{f^2(x)+1} \leq 0$ then find minimum value of $(\mu-\lambda)$ if
$\mathrm{Lim}_{x \rightarrow \lambda^{+}} f(x)=-3$ and $\mathrm{Lim}_{x \rightarrow \mu^{-}} f(x)=3$

(A) 0

(B) 1

(C) -1

(D) None of these

Q.65

AB is a double ordinate of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ such that $\triangle AOB$ (where ' O ' is the origin) is an equilateral triangle, then the eccentricity e of the hyperbola satisfies

(A) $\mathrm{e}>\sqrt{3}$

(B) $1<\mathrm{e}<\frac{2}{\sqrt{3}}$

(C) $e=\frac{2}{\sqrt{3}}$

(D) $e > \frac{2}{\sqrt{3}}$

Q.66

A square matrix P satisfies $P^2 = I - 2P$ where I is the identity matrix if $P^2 + P^3 + P^4 = a^2 I - b^2 P$ then mark incorrect option

(A) $a^2 + b^4$ is a perfect square

(B) $a^4 + b^2$ is a perfect square

(C) $a^2 + b^2$ is prime

(D) $a^4 + b^4$ is prime

Q.67

The area of the region bounded by the curve $C : y = \frac{x+1}{x^2+1}$ and the line $L : y = 1$, is

(A) $1 - \frac{1}{2}\ln 2 + \frac{\pi}{4}$

(B) $\ln 2 - \frac{\pi}{4} + 1$

(C) $\frac{1}{2}\ln 2 + \frac{\pi}{4} - 1$

(D) $\ln 2 - \frac{\pi}{2} + 1$

Q.68

The curve satisfying the differential equation $\frac{dy}{dx} = \frac{y\left(x+y^3\right)}{x\left(y^3-x\right)}$ and passing through $(4, -2)$ is

(A) $y^2 = -2x$

(B) $y = -2x$

(C) $y^3 = -2x$

(D) None of these

Q.69

If the position vectors of the vertices of $\triangle ABC$ are $3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - \hat{\mathbf{k}}, \hat{\mathbf{i}} + 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$ and $5(\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}})$, respectively. Then, the magnitude of the altitude from A onto the side BC is

(A) $\frac{4}{3}\sqrt{5}$

(B) $\frac{5}{3}\sqrt{5}$

(C) $\frac{7}{3}\sqrt{5}$

(D) $\frac{8}{3}\sqrt{5}$

Q.70

Find sum of all possible values of the real parameter ' b ' if the difference between the largest and smallest values of the function $f(x)=x^2-2bx+1$ in the segment $[0,1]$ is 4 .

(A) 5

(B) 1

(C) 3

(D) 2

Q.71

Suppose $f(x)=\left[\begin{array}{ll} x^3 & \text{if } x<1 \\ ax^2+bx+c & \text{if } x\geq 1 \end{array}\right.$. If $f''(1)$ exist then find the value of $a^3+b^3+c^3$.

Q.72

Let $a=3^{\frac{1}{223}}+1$ and for all $n\geq 3$, let $f(n)={}^nC_0\cdot a^{n-1}-{}^nC_1\cdot a^{n-2}+{}^nC_2\cdot a^{n-3}-\ldots\ldots+(-1)^{n-1}\cdot {}^nC_{n-1}\cdot a^0$. If the value of $f(2007)+f(2008)=9k$ where $k\in N$, then find k.

Q.73

Let α, β and γ be three distinct real roots of the equation $x(3x+2)^2+2=(a+12+9x)x^2-bx+c$ where $a, b, c\in R$. If every solution of the inequality $(x-a)^2(4x+b)(x-c)<0$ is also solution of the inequality $3x^2+px+p^2+6p<0$ then find number of integral values of 'p'

Q.74

If $y-\cos x\frac{dy}{dx}=y^2(1-\sin x)\cos x, y(0)=1$, then $y\left(\frac{\pi}{3}\right)=$

Q.75

Let $f: A\to B$ be any function where A is a set containing the positive integral solution of the inequality $\text{cosec}^{-1}(\text{cosec}\,2)>x^2-3x$ and B is the set of all divisors of the natural number 2010. If $f(i)\leq f(j)\ \forall\ i<j$, then find the total number of mappings from A to B.

Answers & Solutions

Q.1 Answer:

—

Solution:

In a transformer

$$\begin{aligned} \frac{N_P}{N_S} &= \frac{I_S}{I_P} \\ \frac{50}{200} &= \frac{I_S}{4} \\ I_S &= 1\ \text{A} \end{aligned}$$

Q.2 Answer:

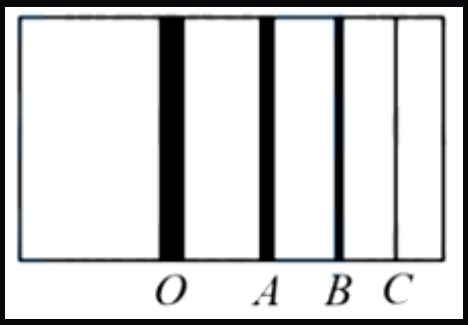
the images C,B and A will shift towards O

Solution:

$b \sin\theta = m\lambda$

$\sin\theta = \frac{m\lambda}{b} \simeq \tan\theta = \frac{y}{D}$

$y \propto \lambda$

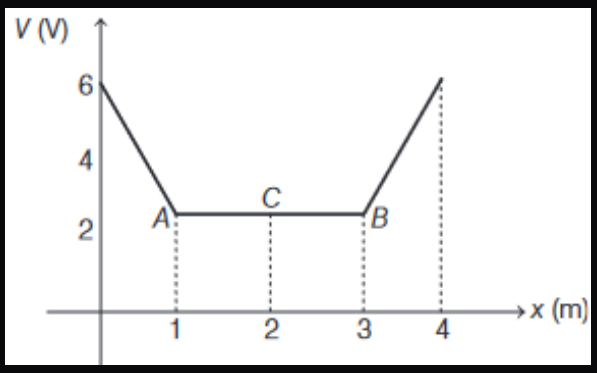


Q.3 Answer:

0

Solution:

According to given diagram, potential is constant in region AB.



i.e., $V =$ **constant**

$\therefore$ Electric field at $x = 2$ m (at point C)

$E = \frac{-dV}{dx} = 0 \quad [\because V = \text{ constant }]$

Q.4 Answer:

0.001 cm

Solution:

Pitch $= \frac{3}{6} = 0.5$ mm

L.C. $= \frac{0.5 \text{ mm}}{50} = \frac{1}{100}$ mm $= 0.01$ mm

$= 0.001$ cm

Q.5 Answer:

4Ω

Solution:

$R_{eq} = R \quad \because 2R \ \& \ 4R$ are short circuited.

$r = R$

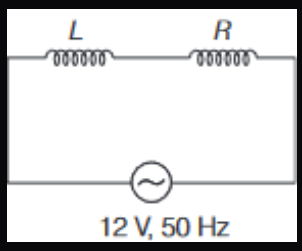
$R = 4\Omega$

Q.6 Answer:

12Ω

Solution:

Given situation following the figure below,



Given for AC circuit,

$V_{\text{rms}} = 12 \text{ V},$

frequency, $f = 50 \text{ Hz},$

current, $I = 0.5 \text{ A},$

phase difference, $\phi = \frac{\pi}{3}$

$\therefore$ Impedance of the AC circuit,

$Z = \frac{V_{\text{rms}}}{I} = \frac{12}{0.5} \Rightarrow Z = 24\Omega$

Power factor,

$\Rightarrow \cos\phi = \frac{R}{Z} \Rightarrow \cos\frac{\pi}{3} = \frac{R}{24} \Rightarrow \frac{1}{2} = \frac{R}{24} \Rightarrow R = 12\Omega$

Q.7 Answer:

$\frac{gT^2}{2}$

Solution:

We know that,

Range of projectile

$R = \frac{u^2 \sin 2\theta}{g}$

As range is maximum, $\theta = 45^\circ$

$R_{\text{max}} = \frac{u^2 \sin 2\times 45}{g} = \frac{u^2}{g}$

Flight time of projectile,

$T = \frac{2u \sin 45^\circ}{g}$

$= \frac{2u}{\sqrt{2} \cdot g} = \frac{\sqrt{2} \cdot u}{g}$

$\text{or} \quad u = \frac{Tg}{\sqrt{2}}$

or

Putting these value of u in Eq. (i), we get

$$R_{\max}=\frac{1}{g}\left(\frac{Tg}{\sqrt{2}}\right)^2$$

$$R_{\max}=\frac{1}{2}gT^2$$

Q.8 Answer:

$\left(2\times10^{-7}\right)\sin[200\pi(ct-x)]\hat{k}T$

Solution:

Equation of magnetic field of an electromagnetic wave is given by

$B=B_0\sin(\omega t-kx)\hat{k}$

$=B_0\sin\left(2\pi ft-\frac{2\pi}{\lambda}\cdot x\right)\hat{k}\quad[\because\omega=2\pi f$ **and** $k=\frac{2\pi}{\lambda}]$

$=B_0\sin\frac{2\pi}{\lambda}[f\lambda t-x]\hat{k}$

$B=B_0\sin\frac{2\pi}{\lambda}[ct-x]\hat{k}$ **...(i)**

Also direction of $\mathbf{B}$ **vector is perpendicular to** $\mathbf{E}$ **and also to direction of propagation**

And $B_0=\frac{E_0}{c}$

Here, $E_0=60\mathrm{Vm}^{-1}$

$\therefore\quad B_0=\frac{60}{3\times10^8}=2\times10^{-7}$

Also $k=\frac{2\pi}{\lambda}$

Here $k=200\pi=\frac{2\pi}{\lambda}$

$\Rightarrow\quad\lambda=\frac{1}{100}\,\mathrm{m}=\frac{100}{100}\,\mathrm{cm}=10\,\mathrm{mm}=10^{-2}\,\mathrm{m}$

Putting the values is eq. (i), we get

$B=2\times10^{-7}\sin\left[\frac{2\pi}{10^{-2}}(ct-x)\right]\hat{k}$

$=2\times10^{-7}\sin200\pi(t-x)\hat{k}$

Q.9 Answer:

Electron will keep orbiting in the ground state of the atom

Solution:

The energy of the photon

$$E=\frac{hC}{\lambda}$$

$$=\frac{4\times10^{-15}\mathrm{eVs}\times3\times10^8\mathrm{m/s}}{300\times10^{-9}\mathrm{m}}$$

$$=\frac{4\times10^2}{300}\mathrm{eV}=\frac{4}{3}\mathrm{eV}=1.33\mathrm{eV}$$

The ionisation energy is $13.6\mathrm{eV}$ **which is greater than energy of photon, so atom can not come into excited state and will remain in ground state**

Q.10 Answer:

$\frac{b}{c}$

Solution:

Given, $y = a\sin(bt - cx)$

Comparing the given equation with general wave equation

$y = a\sin\left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda}\right)$

we get

$b = \frac{2\pi}{T}, c = \frac{2\pi}{\lambda}$

(a) Dimensions o

$$\text{of } \frac{y}{a} = \frac{\text{metre}}{\text{metre}} = \frac{[\mathrm{L}]}{[\mathrm{L}]}$$
$$= \text{ Dimensionless}$$

(b)

$$\text{Dimensions of } bt = \frac{2\pi}{T} \cdot t = \frac{[\mathrm{T}]}{[\mathrm{T}]}$$
$$= \text{ Dimensionless}$$

(c) Dimensions of

$$cx = \frac{2\pi}{\lambda} \cdot x = \frac{[\mathrm{L}]}{[\mathrm{L}]}$$
$$= \text{ Dimensionless}$$

(d) Dimensions of

$$\frac{b}{c} = \frac{2\pi}{T} / \frac{2\pi}{\lambda}$$
$$= \lambda/T = \left[\mathrm{LT}^{-1}\right]$$

Thus, option (d) has dimensions.

Q.11 Answer:

H

Solution:

According to law of conservation of energy at point B shown in the figure given in question part,

Loss in PE = Gain in KE

$\Rightarrow \quad mg(H-h) = \frac{1}{2}mv^2$

$\Rightarrow \quad v = \sqrt{2g(H-h)}$

Now, $\quad h = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2h}{g}}$

$\therefore$ Distance covered in horizontal portion,

$s = v \times t$

$= \sqrt{2g(H-h)} \times \sqrt{\frac{2h}{g}}$

$\Rightarrow \quad s = \sqrt{4h(H-h)}$ ...(i)

For maximum value of s

$\frac{ds}{dh} = 0$

$\Rightarrow \quad 2(H-2h) = 0 \Rightarrow H = 2h$

$\Rightarrow \quad h = \frac{H}{2}$

Substituting the value of h in Eq. (i), we get,

$$s = \sqrt{4\frac{H}{2}\left(H - \frac{H}{2}\right)}$$

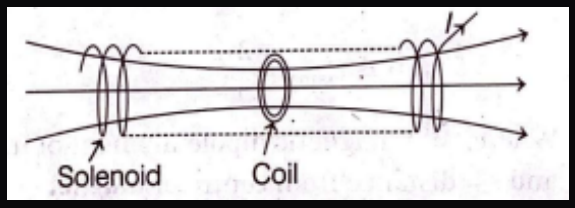
$$= \sqrt{2H \cdot \frac{H}{2}} = H$$

Q.12 Answer:

$0.4\ \mathrm{V}$

Solution:

Due to solenoid a magnetic field exists around coil placed at centre



Magnetic field intensity produced by solenoid,

$B = \mu_0 nI$

Flux linked with coil is

$\phi_B = N.B \cdot A$

Where, N = Number of turns in coil

A = Area of coil

As current in solenoid is reversed, change in flux is

$\Delta\phi_B = 2N \cdot B \cdot A$

This change in flux produces an emf in coil given by

Induced emf e = Rate of change of flux

$\Rightarrow \quad e = \frac{2NBA}{\Delta t} = \frac{2N(\mu_0 nI)A}{\Delta t}$

Here $N = 200, A = 25\ \mathrm{cm}^2 = 25 \times 10^{-4}\ \mathrm{m}^2$

$I = \frac{4}{\pi}\mathrm{A}, n = 100$ **tuns** $/\mathrm{cm} = 100 \times 100$ **turns** $/\mathrm{m}$

$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{T \cdot mA^{-1}}$ **and** $\Delta t = 0.04\ \mathrm{s}$

So induced emf,

$e = \frac{2\times200\times4\pi\times10^{-7}\times10^{4}\times\frac{4}{\pi}\times25\times10^{-4}}{0.04} = 0.4\ \mathrm{V}$

Induced emf in coil is $0.4\ \mathrm{V}$

Q.13 Answer:

A, B true, but C false

Solution:

In hydrogen atom, potential energy of electron in nth orbit is given as

$\mathrm{PE} = \frac{-Kze^2}{r}$

For higher value of n, r increase because $r \propto n^2$.

From eq (i), it is clear that for higher value of n, potential energy increases.

Hence, statement (A) is correct.

In thomson's model an atom is a spherical cloud of positive charges with electrons embedded in it.

Hence, statement (B) is correct.

According to Bohr's model an electron in an atom is located at a definite distance from the nucleus and is revolving with a definite velocity around it. According to heisenberg's uncertainty principle, it is impossible to determine the exact position and momentum of moving electron simultaneously.

Therefore, Bohr's model of an atom is contradicted by heisenbergs uncertainty principle.

The statement (C) is incorrect.

Hence, option (b) is correct.

Q.14 Answer:

4

Solution:

$n = 5$ moles, $T_1 = 100^\circ\text{C}$

$T_2 = 120^\circ\text{C}, \Delta U = 80\ \text{J}$

Rise in temperature $\Delta t = 120 - 100 = 20^\circ\text{C}$

$\Delta U = ms\Delta t$

$\frac{80}{5} = 1 \times s \times 20$

$s = 0.8\ \text{J}$

$\therefore$ For 5 moles, $s = 0.8 \times 5\ \text{J/K}$

$= 4\ \text{J/K}$

Q.15 Answer:

$\frac{100}{9}$

Solution:

Resistance of galvanometer

$$G = 100\Omega$$

$$I_g = 100\mu\text{A} = 100 \times 10^{-6}\ \text{A} = 0.1 \times 10^{-3}\ \text{A}$$

$$I = 1\ \text{mA} = 1 \times 10^{-3}\ \text{A}$$

Shunt resistance

$$S = \frac{I_g G}{I - I_g} = \frac{0.1 \times 10^{-3} \times 100}{1 \times 10^{-3} - 0.1 \times 10^{-3}}$$

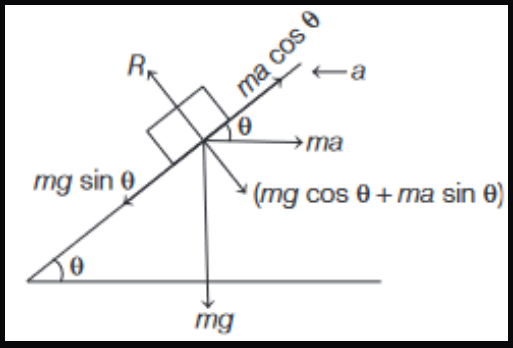
$$= \frac{1 \times 10^{-2}}{0.9 \times 10^{-3}} = \frac{10^2}{9} = \frac{100}{9}\Omega$$

Q.16 Answer:

$mg \sec\theta$

Solution:

The given situation is shown in the following figure,



The acceleration given to wedge to the left is a.

The block has a pseudo acceleration to the right, pressing against the wedge because of which the block does not move.

In this case,

$$\therefore \quad mg\sin\theta = ma\cos\theta$$

$$\Rightarrow \quad a = \frac{g\sin\theta}{\cos\theta} \quad \ldots(i)$$

Force exerted by the wedge on the block is equal to normal reaction on the block,

$$\text{i.e. } R = mg\cos\theta + ma\sin\theta$$

$$= mg\cos\theta + m\cdot g\frac{\sin\theta}{\cos\theta}\cdot\sin\theta \quad [\text{From Eq. (i)}]$$

$$= \frac{mg}{\cos\theta}\left[\cos^2\theta + \sin^2\theta\right]$$

$$\Rightarrow R = \frac{mg}{\cos\theta} = mg\sec\theta$$

Q.17 Answer:

$\frac{2000}{3}\Omega$

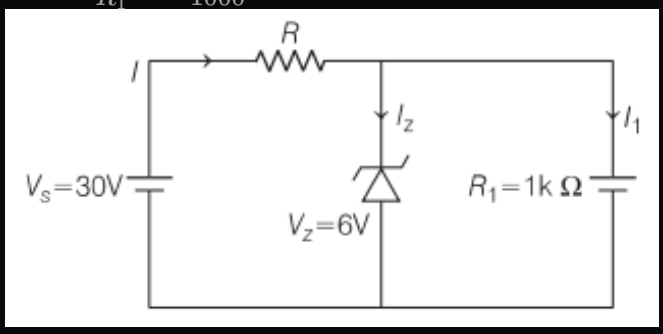
Solution:

Given that, $R_1 = 1\text{k}\Omega$,

Zener breakdown voltage $V_z = 6$ V

Let I_1 and I_z be the current passing through R_1 and Zener diode, then using equation,

$$I_1 = \frac{V_z}{R_1} = \frac{6}{1000}\text{ A}$$



According to given condition Zener current,

$$I_z = 5I_1 = \frac{30}{1000}\text{ A}$$

Let I be the current drawn from source battery $(V_s = 30\text{ V})$, then

$$I = \frac{V_s - V_z}{R} = I_1 + I_z$$

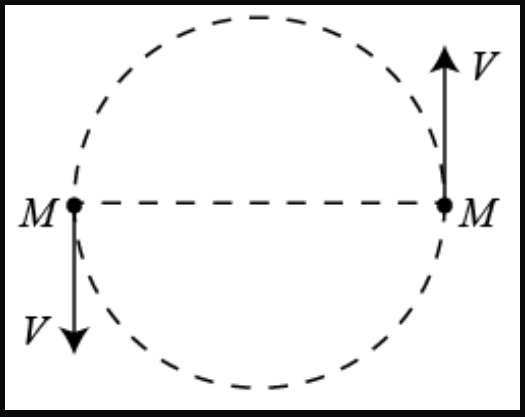
$\frac{30-6}{R} = \frac{6}{1000} + \frac{30}{1000}$

$\Rightarrow \quad R = \frac{2000}{3}\Omega$

Q.18 Answer:

momentum changes by $2MV$

Solution:



$$P_i = +MV$$
$$P_f = -MV$$
$$\Delta P = 2MV$$

Q.19 Answer:

$1200\ \text{ms}^{-1}$

Solution:

$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$

Given, at temperature T_1,

$$v_{\text{rms}} = 600 = \sqrt{\frac{3RT_1}{M}} \qquad \ldots(\text{i})$$

At temperature, $T_2 = 2\pi T_1$,

due to dissociation

$$v_{\text{rms}} = v = \sqrt{\frac{3R(2T_1)}{(M/2)}} \qquad \ldots(\text{ii})$$

Dividing Eq. (i) by Eq. (ii), we get

$\Rightarrow \quad \frac{v}{600} = 2 \quad \Rightarrow v = 1200\ \text{ms}^{-1}$

Q.20 Answer:

$\left[\frac{2G(m_1+m_2)}{r}\right]^{1/2}$

Solution:

Initially when the two masses are at an infinite distance from each other, their gravitational potential energy is zero. When they are at a distance r from each, the gravitational PE is

$\mathrm{PE} = \frac{-Gm_1m_2}{r^2}$

The minus sign-indicates that there is a decrease in PE. This gives rise to an increase in kinetic energy. If v_1 and v_2 are their respective velocities when they are a distance r apart then, from the law of conservation of energy, we have

$\frac{1}{2}m_1v_1^2 = \frac{Gm_1m_2}{r}$ or $v_1 = \sqrt{\frac{2Gm_2}{r}}$

and $\frac{1}{2}m_2v_2^2 = \frac{Gm_1m_2}{r}$ or $v_2 = \sqrt{\frac{2Gm_1}{r}}$

Therefore, their relative velocity of approach is

$$v_1 + v_2 = \sqrt{\frac{2Gm_2}{r}} + \sqrt{\frac{2Gm_1}{r}} = \sqrt{\frac{2G}{r}(m_1 + m_2)}$$
$$= \left(\frac{2G(m_1 \times m_2)}{r}\right)^{\frac{1}{2}}$$

Q.21 Answer:

—

Solution:

$\mathrm{C_s} = 20\ \mathrm{F}, V = 5$ volt,

$F_x = 30\ \mathrm{F}$

Decrease in energy,

$\Delta U = \frac{1}{2}\frac{C_1C_2}{C_1+C_2}(V_2 - V_2)^2$

$= \frac{1}{2} \times \frac{20 \times 30}{20 + 30}(5 - 0)^2$

$= \frac{300}{50} \times 25 = 150\ \mathrm{J}$

Q.22 Answer:

—

Solution:

We know that Intensity of magnetisation

$I = \frac{M}{V}$

[where M = magnetic moment, V = volume]

So

$$M = IV = 5.0 \times 10^4 \times \frac{12}{100} \times \frac{1}{(100)^2}$$
$$= 60 \times 10^4 \times 10^{-6} = 0.6 \mathrm{Am}^2$$

Q.23 Answer:

—

Solution:

When atom are excited from ground state to excited state with $n = 20$, then number of spectral line possible are
$= \frac{n(n-1)}{2}$ **[where, n = principal quantum number]**
$= \frac{20(20-1)}{2} = 190$

Q.24 Answer:

—

Solution:

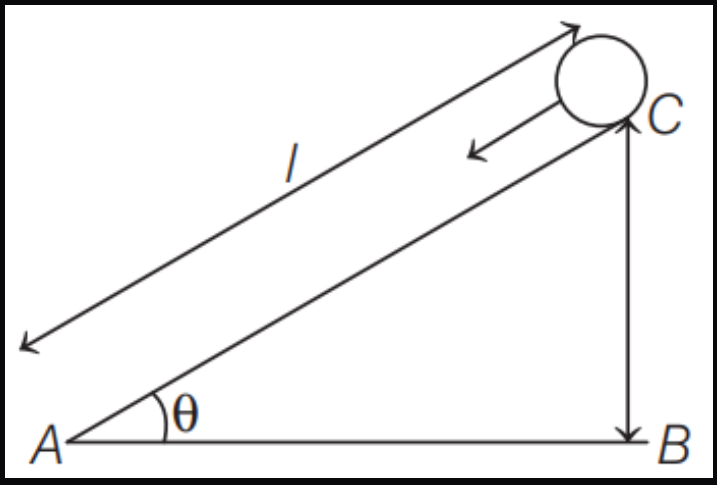
Distance travelled in first 2 seconds $= 200 \text{ m}$
$S = ut + \frac{1}{2}at^2$
$200 = 2u + \frac{1}{2} \times a \times 2^2$
$u + a = 100$ **...(1)**
Distance travelled in next 4 seconds $= 220 \text{ m}$ **Total distance travelled in first 6 second** $= 420 \text{ m}$
$420 = 6u + \frac{1}{2} \times 9 \times 6^2$
$420 = 6u + 18a$
$\mathrm{u} + 3\mathrm{a} = 70$ **...(2)**
Solving equation (1) and (2), we have
$\mathrm{a} = -15 \text{ m/s}^2, \mathrm{u} = 115 \text{ m/s}$
$\mathrm{v} = \mathrm{u} + \mathrm{at}$
$= 115 - 15 \times 7$
$= 115 - 105 = 10 \text{ m/s}$

Q.25 Answer:

—

Solution:

According to the question, given situation is shown below



In $\triangle ABC$,

$\sin\theta = \frac{BC}{AC} \Rightarrow BC = AC\sin\theta = l\sin\theta$

From law of conservation of mechanical energy,

$mg(l\sin\theta) = \frac{1}{2}I\omega^2 + \frac{1}{2}mv^2$

$mg(l\sin\theta) = \frac{1}{2}\left(\frac{1}{2}mr^2\right)\left(\frac{v}{r}\right)^2 + \frac{1}{2}mv^2$

$\left[\because I = \frac{1}{2}mr^2 \text{ and } \omega = \frac{v}{r}\right]$

$mgl\sin\theta = \left(\frac{1}{4} + \frac{l}{2}\right)mv^2$

$gl\sin\theta = \frac{3}{4}v^2 \Rightarrow v^2 = \frac{4gl\sin\theta}{3}$

$\Rightarrow \quad v = \sqrt{\frac{4gl\sin\theta}{3}} = \sqrt{\frac{4\times10\times0.6\times\sin30^\circ}{3}}$

$[\because l = 60\text{ cm} = 0.6\text{ m}]$

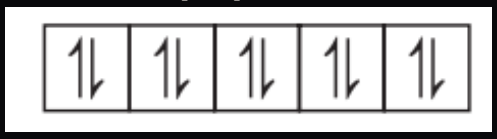
$= \sqrt{\frac{12}{3}} = \sqrt{4} = 2\text{ m/s}$

Q.26 Answer:

Aqueous solutions of Cu^+ and Zn^{2+} are colourless

Solution:

(a) $Cu^+ = [Ar]3d^{10}$



(No unpaired electrons, so it is colourless)

$Zn^{2+} = [Ar]3d^{10}$

(No unpaired electrons, so it is colourless)

(b) $Cu^{2+} = [Ar]3d^9$



(One unpaired electron is present, so it is coloured) Zn^{2+} **is colourless.**

(c) The colour of Fe^{2+} **is green while that of** Fe^{3+} **is blue.**

(d) MnO_4^- contains no unpaired electrons, however, it is coloured due to charge transfer.

Thus, only statement given in option (a) is correct.

Q.27 Answer:

2 4 1 5

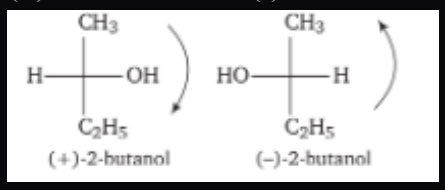
Solution:

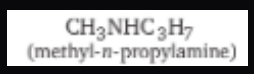


Thus, these two are tautomers.

Eclipsed and staggered ethane are two conformations of ethane.

(+)2-butanol and (-)2-butanol are enantiomers as these are non-superimposable mirror images.







Due to difference in the nature of alkyl groups attached to the same functional group, these are called metamers.

Q.28 Answer:

$s > p > d > f$

Solution:

s has the greatest effect to reduce the force of attraction between the outermost electron and nucleus due to its effective charge density followed by $p > d > f$.

So, according to the screening effect, s orbital being close to the nucleus lacks screening from any other orbital as it is the first and the closest orbital. p orbital electrons are loosely bound with greater size than s block elements as they undergo screening from s orbital electrons.

Similarly, d-orbital electrons undergo screening from s and p-block orbitals and finally f orbital also undergoes screening from all the other orbitals. Hence, the f-block elements are generally big in size with less nuclear charge acting on their outermost electrons.

Thus, the order of screening effect of electrons of s, p, d and f orbitals of a given shell of an atom on its outer shell electrons is $s > p > d > f$.

Hence, the correct option is (1).

Q.29 Answer:

SO_3^{2-}, SO_2

Solution:

According to question, $[X]+$ **Dil.** $H_2SO_4 \longrightarrow [Y]$ **:**

Colourless, suffocating gas $[Y] + K_2CrO_7 + H_2SO_4 \longrightarrow$ **Green colouration of solution.**

All sulphites when treated with dil. H_2SO_4 **giver colourless and suffocating sulphur dioxide gas.**

$$SO_3^{2-}(s) + \text{Dil} \cdot H_2SO_4(aq) \longrightarrow SO_4^{2-}(aq) + SO_2(g)$$

$$+H_2O(l)$$

SO_2 **gas** $[Y]$ **turns acidified potassium dichromate paper green due to reduction of** Cr **(VI) to Cr (III).**

$$K_2Cr_2O_7(aq) + H_2SO_4(aq) + 3SO_2(g) \longrightarrow$$

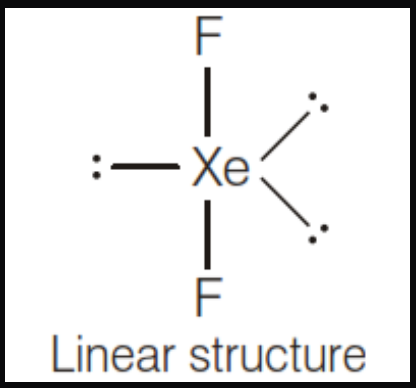
$$K_2SO_4(aq) + Cr_2(SO_4)_3(aq) + H_2O(l)$$

Q.30 Answer:

(A) is false, but (R) is true

Solution:

XeF_2 **is** sp^3d **hybridised.** XeF_2 **molecule does not follow octet rule due to expanded octet 10 electrons. Therefore,** A **is incorrect but** R **is correct.**



Q.31 Answer:

Cellulose is not digestible in human body

Solution:

Cellulose is a polysaccharide which holds or binds many small monosaccharide units together by glycosidic bonds. The type of glycosidic bond present is of $(\beta) - 1,4$ **bond. Which cannot be broken down by enzymes present in human body. Thus, we cannot digest cellulose.**

Q.32 Answer:

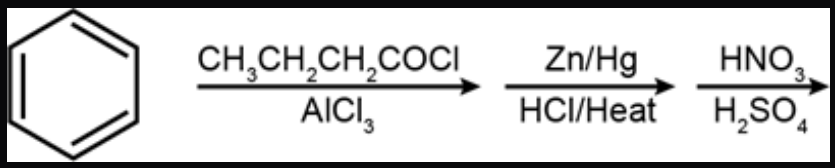
liquid solvent and solid solvent

Solution:

Freezing point of a substance is the temperature at which the solid and the liquid forms of the substance are in equilibrium.

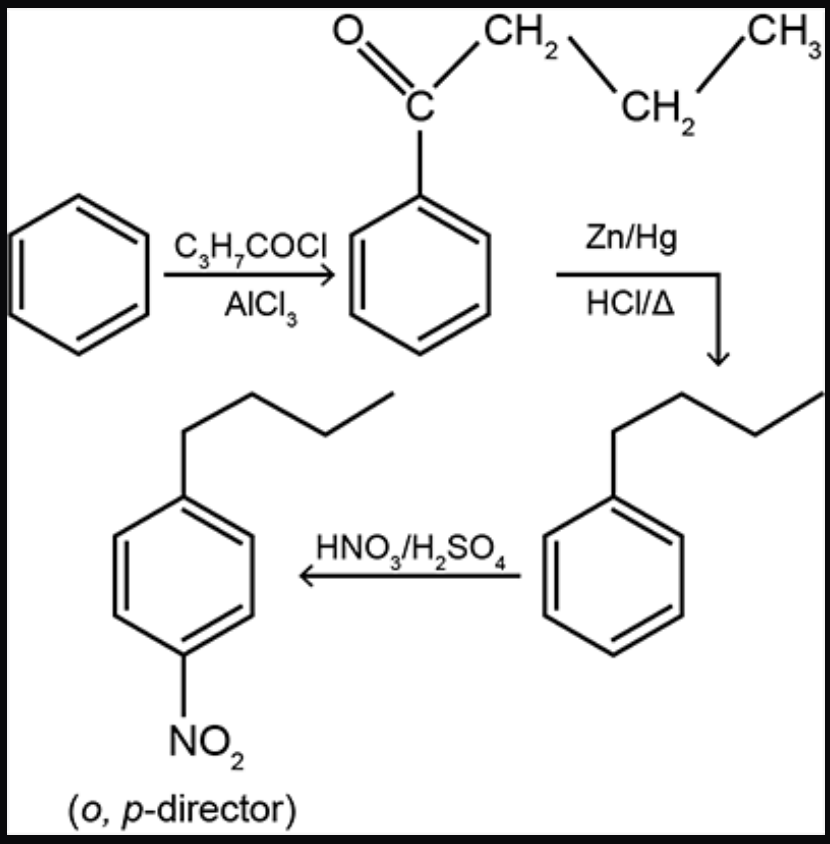
If a non-volatile solute is added to the solvent, there is decrease in vapour pressure of the solution and thus the freezing point of the solution is less than that of pure solvent. It is called depression in freezing point.

Q.33 Answer:



Solution:

The sequence of reaction is as follows:



Q.34 Answer:

(ii) (iii) (iv) (v)

Solution:

I. $[\mathrm{Co}\,(\mathrm{NH_3})_6]^{3+}$

Here Co is present as Co^{3+} ion. $\mathrm{NH_3}$ being strong field ligand pair up the unpaired electrons of Co.

II.$[\mathrm{Ni(CO)_4}]$

Here Ni is present in its ground state. CO being a strong field ligand, pair up the unpaired electrons of Ni

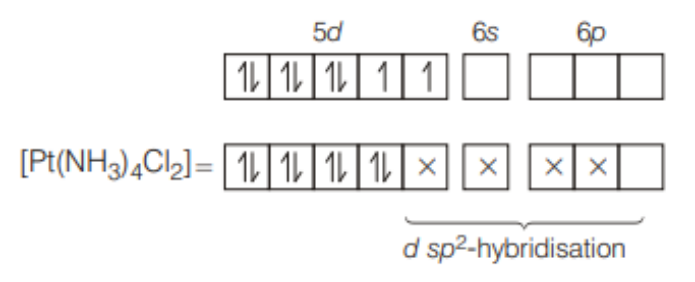
$_{28}\mathrm{Ni} = [\mathrm{Ar}]3d^8 4s^2$

III. $[\mathrm{Pt}\,(\mathrm{NH_3})_2\,\mathrm{Cl_2}]$

Here Pt is present as Pt^{2+} ion.

$_{78}Pt = [Xe]5d^9 6s^1$

$Pt^{2+} = [Xe]5d^8 6\ s^0$



IV. $[CoF_6{}^{\beta-}]$

Here Co is present as Co^{3+} ion, F being weak field ligand is unable to pair up its unpaired electrons.

$Co^{3+} = [Ar]3d^6 4s^0$

V. $[Fe(CO)_5]$

Here Fe is present in its ground state.

$_{26}Fe = [Ar]3d^6 4s^2$

Q.35 Answer:

Lu^{3+}

Solution:

Generally, paramagnetism is shown by lanthanoid ions. The paramagnetism arises due to presence of unpaired electrons in f-orbital.

$Lu^{3+}\ (4f^{14})$ does not have unpaired electrons.

Hence, it does not exhibit paramagnetism.

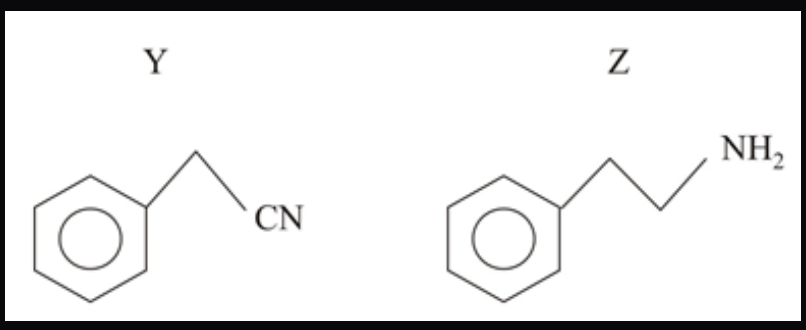
Q.36 Answer:

Both (A) and (R) are true but (R) is not the correct explanation of (A).

Solution:

In cycloalkane the heat of combustion of cyclohexane $(657.9\ kJ\ mol^{-1})$ is the lowest hence it is the most stable cycloalkane. Cyclopropane and cyclobutane are less stable due to angle strain and torsional strain, therefore cyclopropane and cyclobutane are reactive cycloalkane.

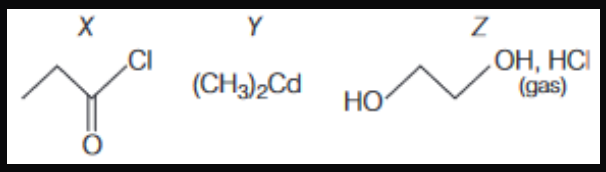
Q.37 Answer:



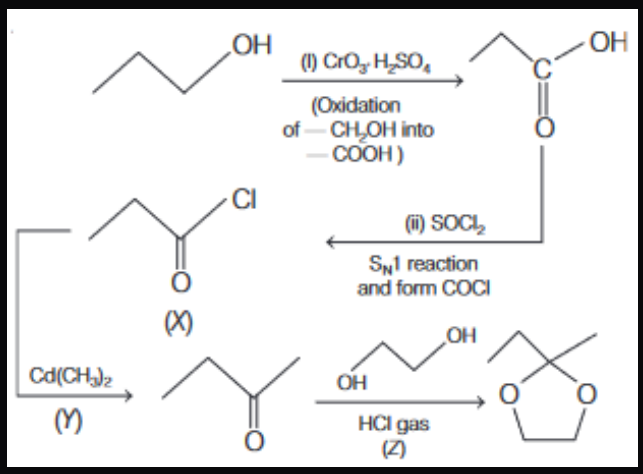
Solution:

An organic compound $X(C_7H_7Cl)$ when reacted with KCN/C_2H_5OH gave a major product Y. Z is formed when Y is reduced with $LiAlH_4$. What are Y and Z?

Q.38 Answer:



Solution:



Hence, option (3) is correct.

Q.39 Answer:

16

Solution:

Maximum number of electrons

$= 2n^2 = 2 \times 4 \times 4 = 32$(when $n = 4$)

Half of the electrons have $\left(+\frac{1}{2}\right)$ spin, i.e., 16

Q.40 Answer:

$C_2H_5 - CO - C_2H_5$

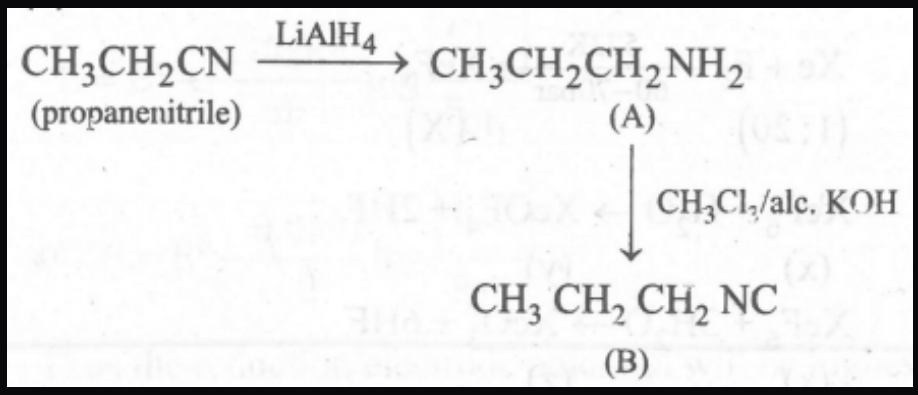
Solution:

Iodoform test is used to identify methyl ketones. It is used for the detection of ketones and aldehydes having an alpha methyl group. As, option (c) does not have any methyl group, it does not respond to iodoform test.

Q.41 Answer:

Carbyl amine reaction

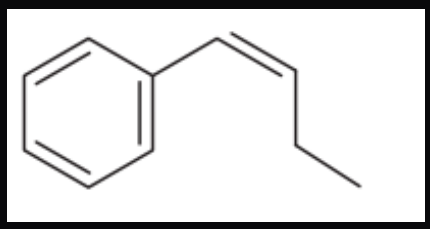
Solution:



Product ' B ' is an isocyanide or carbylamine that has an offensive smell.

The reaction is called Carbylamine Test or_Carbylamine reaction and is used to defect primary amines.

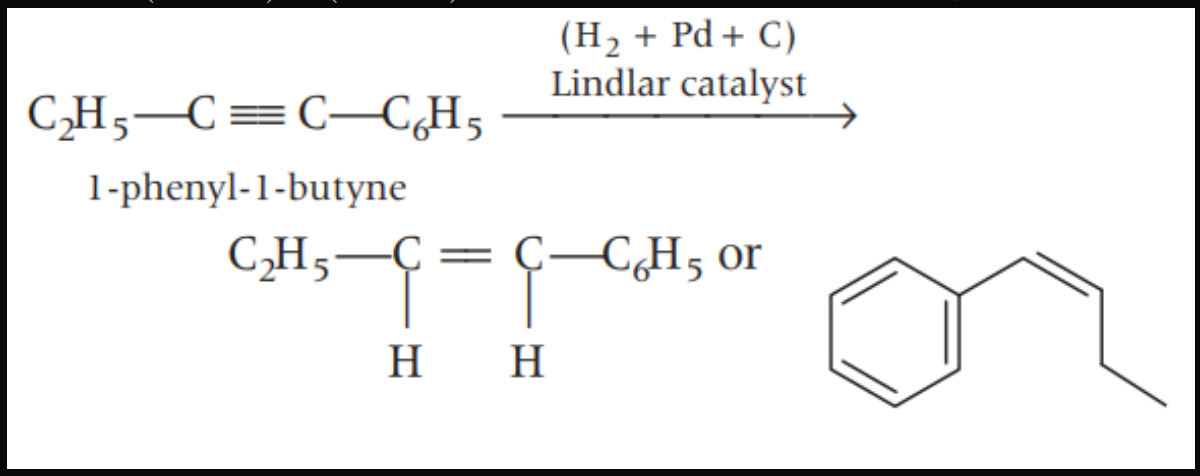
Q.42 Answer:



Solution:

Lindlar catalyst is partially deactivated catalyst composed of $BaSO_4$ coated with Pd poisoned with quinoline.

It reduce $(C \equiv C)$ to $(C = C)$ bond, occurs via syn-addition and gives cis-alkene.



Q.43 Answer:

b, c, d, e

Solution:

b,c,d,e follow huckel rule so these are aromatic

a is anti aromatic

Q.44 Answer:

II, III, IV only

Solution:

The crystal filed theory is successful in explaining which of the following?
I. Ligands as point charges;
II. Formation and structures of complexes.
III. colour;
IV. Magnetic properties:
V . covalent character of metal-ligand bonding.

Q.45 Answer:

Both (A) and (R) are correct but (R) is not the correct explanation of (A)

Solution:

SiF_6^{2-} is formed but $SiCl_6^{2-}$ is not because Cl has greater size than F. Therefore, larger size of anions cannot form complex with silicon.
Electronegativity of F is higher than Cl but cannot explain about Assertion.
Thus, Assertion and Reason both are correct but Reason (R) is not the correct explanation of (A).

Q.46 Answer:

—

Solution:

$$\mathrm{W} = -\mathrm{P}_{\text{ext}}\left(\mathrm{V}_2 - \mathrm{V}_1\right) = -2(\mathrm{x} - 5) = -2026.4\ \mathrm{J}$$
$$= -20\ \mathrm{L.atm}$$
$$\Rightarrow -2x + 10 = -20 \Rightarrow 2x - 10 = 20$$
$$\Rightarrow x = 15L$$

Q.47 Answer:

—

Solution:

Valency of metal,

$$(n) = \frac{2 \times \text{ vapour density}}{\text{Equivalent weight of metal } + 35.5}$$
$$= \frac{2 \times 83}{6 + 35.5} = 4$$

The atomic weight of metal is equal to the product of the equivalent weight of metal and valency.
Atomic weight of metal $= n\times$ equivalent weight of metal
$= 4 \times 6 = 24$

Q.48 Answer:

—

Solution:

The molecular orbital electronic configuration of O_2 is
$O_2\left(8+8=16e^{-}\right)=\sigma 1s^2, \sigma 1s^2, \sigma 2s^2, \sigma^* 2s^2,$
$\sigma 2p_z^2, \pi 2p_x^2, \approx \pi 2p_y^2, \overset{*}{\pi}\, 2p_x^1 \approx \overset{*}{\pi}\, 2p_y^1$

(Unstarred σ and π represent bonding orbitals.)
$\therefore$ number of bonding electrons $= 10$
and number of bonding electron pairs $= 5$

Q.49 Answer:

—

Solution:

$Cr_2O_7^{2-} + Fe^{2+} \rightarrow Fe^{3+} + Cr^{3+}$
n factor of $Cr_2O_7^{2-} = 6$
n factor of $Fe^{2+} = 1$
So to reduce one mole of dichromate 6 moles of Fe^{2+} is required.

Q.50 Answer:

—

Solution:

For 1 mole of $\left(Mg \longrightarrow Mg^{2+}\right)$ IE $= \mathbb{E}E_1 + IE_2 = (740 + 1450) = 2190$ kJ/mol.
Number of mole in 4.8 g of $Mg = 4.8/24 = 0.2$ mol
For 1 mole energy required $= 2190$ kJ/mol
For 0.2 mole energy required $= 2190 \times 0.2 = 438$ kJ
Thus, for 4.8 g of Mg to Mg^{2+} conversion, energy required is 438 kJ.

Q.51 Answer:

0.75

Solution:

$N = 48$
$\frac{N}{2} = 24$
$\therefore$ Median $= 24$ th observation $= 12$

x_i	f_i	cf	$\lvert x_i - M\rvert$	$f_i \lvert x_i - M\rvert$
10	6	6	2	12
11	12	18	1	12
12	18	36	0	0
13	12	48	1	12

$N = 48$

$$\text{MD} = \frac{\Sigma f_i \left| x_i - M \right|}{N} = \frac{36}{48} = 0.75$$

Q.52 Answer:

6

Solution:

Let first term and common ratio be a and r respectively.

Now, $\sum_{n=1}^{100} g_{2n} = ar + ar^3 + ar^5 + \ldots\ldots\ldots + ar^{199} = \frac{10}{3}$ **...(1)**

Also, $\sum_{n=1}^{100} g_{2n-1} = a + ar^2 + ar^4 + \ldots\ldots\ldots + a^{198} = \frac{5}{9}$ **...(2)**

Now, $\frac{\text{Equation (1)}}{\text{Equation (2)}} \Rightarrow \frac{r\left(a+a^2+\ldots\ldots\ldots+\text{ar}^{198}\right)}{(\text{a}+\text{ar}^2+\ldots\ldots\ldots+\text{ar}^{198})} = \frac{10/3}{5/9} \Rightarrow r = 6$

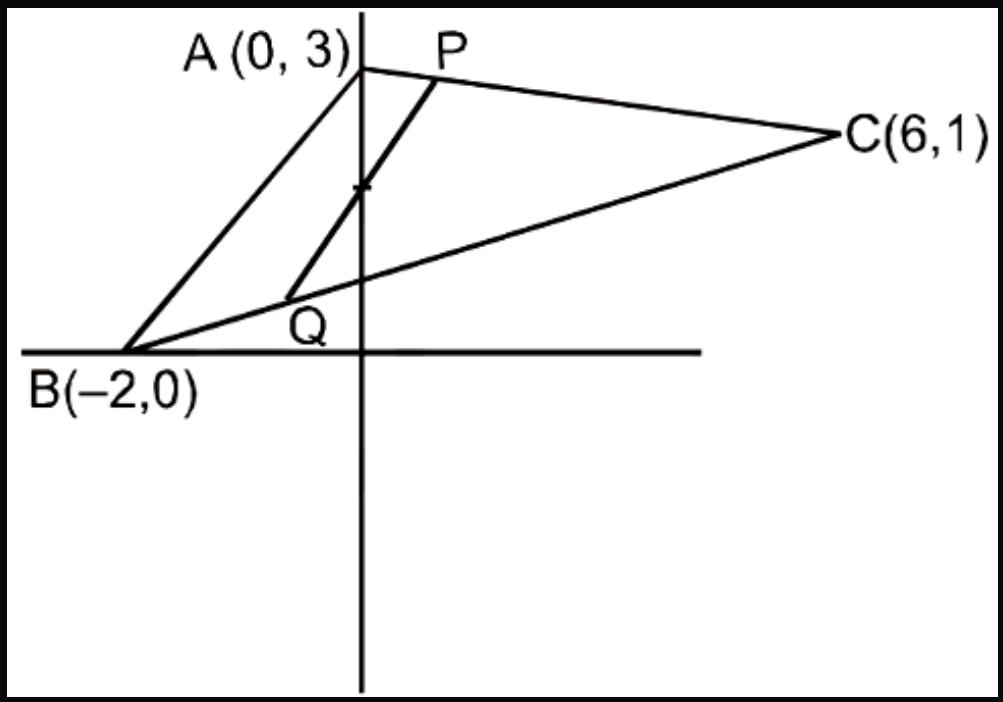
Q.53 Answer:

2

Solution:

Since $(\lambda, \lambda + 1)$ **lies on** $y = x + 1$

equation of $AB : 3x - 2y + 6 = 0; BC : x - 8y + 2 = 0; AC : x + 3y - 9 = 0$



Line $y = x + 1$ **cuts** AC **at** $P\left(\frac{3}{2}, \frac{5}{2}\right)$ **cut** BC **at** $Q\left(\frac{-6}{7}, \frac{1}{7}\right)$**. Hence** $\lambda \in \left(\frac{-6}{7}, \frac{3}{2}\right)$

Q.54 Answer:

288

Solution:

$\because \vec{v}_1 \perp \vec{v}_2$

$\Rightarrow a_1a_2 + b_1b_2 + c_1c_2 = 0$

Case-1 $a_1a_2 = 4 \Rightarrow (a_1, a_2) = (-2,-2), (2,2)$

$\therefore$ blb2 $= -2 \Rightarrow (b_1, b_2) \equiv (1,-2), (-1,2), (2,-1),$
$(-2,1)$

& $c_1c_2 = -2 \Rightarrow (c_1, c_2)$ **has four cases**

$\therefore$ **total number of cases** $= 2 \times 4 \times 4 \times 3 = 96$

Case-2 $a_1a_2 = -4 \Rightarrow (a_1, a_2) = (-2,2)$ **or** $(2,-2)$ $b_1\ b_2 = 2 \Rightarrow (b_1,\ b_2) =$
$(1,2), (2,1), (-1,-2), (-2,-1)$

Similarly $c_1c_2 = 2$ **gives 4 cases**

$\therefore$ total number of cases $= 2 \times 4 \times 4 \times 3 = 96$

Case-3 $a_1a_2 = 2$

$\Rightarrow (a_1, a_2) \equiv (2,1), (1,2), (-2,-1), (-1,-2)$

$b_1b_2 = -1 \Rightarrow (b_1, b_2) = (1,-1), (-1,1)$

$c_1c_2 = -1 \Rightarrow$ Two cases

$\therefore$ total number of cases $= 4 \times 2 \times 2 \times 3 = 48$

Case-4 $a_1a_2 = -2$

$\Rightarrow (a_1, a_2) \equiv (1,-2), (-1,2), (2,-1), (-2,1)$

$b_1\ b_2 = 1 \Rightarrow (b_1,\ b2) \equiv (-1,-1), (1,1)$&

$c_1c_2 = 1 \Rightarrow$ Two cases

$\therefore$ total number of cases $= 48$

total number of cases $= 96 + 96 + 48 + 48 = 288$

Q.55 Answer:

$\frac{22}{\sqrt{17}}$

Solution:

Given, lines $\mathbf{r} = (-\hat{\mathbf{i}} + 3\hat{\mathbf{k}}) + t(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}})$

and

$\mathbf{r} = (3\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}) + s(2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}})$

Shortest distance between lines is

$$\text{SD} = \frac{((3+1)\hat{\mathbf{i}} + \hat{\mathbf{j}} - (\hat{\mathbf{k}} + 3)\hat{\mathbf{k}}) - [(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}}) \times (2\mathbf{i} - \mathbf{j} + 2\mathbf{k})]}$$

$$\text{SD} = \frac{(4\hat{\mathbf{i}} + \hat{\mathbf{j}} - 4\hat{\mathbf{i}}) \cdot (12\hat{\mathbf{j}} + 6\hat{\mathbf{i}})| \times |(2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{j}} - 8\hat{\mathbf{k}})\ |}{|12\hat{\mathbf{i}} + 8\hat{\mathbf{j}} - 8\hat{\mathbf{k}}|}$$

$$\text{SD} = \frac{48 + 8 + 32}{\sqrt{144 + 64 + 64}} = \frac{88}{4\sqrt{17}} = \frac{22}{\sqrt{17}}$$

Q.56 Answer:

$(-1, 5)$

Solution:

$\text{Lim}_{n\to\infty} \frac{1}{\frac{(x-2)^n}{3^n}+3-\frac{1}{n}}$

for $\text{Lim}_{n\to\infty}$ **to be equal to** $\frac{1}{3}$

$\text{Lim}_{n\to\infty} \frac{1}{n} \to 0$ **which is** T **and** $\text{Lim}_{n\to\infty} \left(\frac{x-2}{3}\right)^n \to 0$

$$-1 < \frac{x-2}{3} < 1$$
$$\Rightarrow -3 < (x-2) < 3$$
$$\Rightarrow -1 < x < 5$$
$$\Rightarrow (-1, 5)$$

Q.57 Answer:

$\frac{27}{128}$

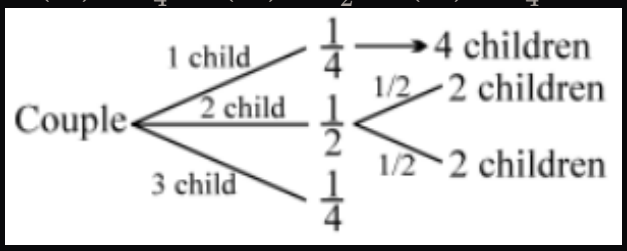
Solution:

A : exactly one child

B **: exactly two children**

C **: exactly 3 childre**

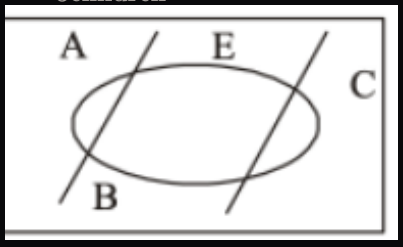
$P(A) = \frac{1}{4}; P(B) = \frac{1}{2}; P(C) = \frac{1}{4}$



E : couple has exactly 4 grandchildren

[OPTION 1]$P(E) = P\left[OPTION2\right] \cdot P(E/A) + P\left[OPTION3\right] \cdot P(E/B) + P \cdot P(E/C)$

$$= \underbrace{\frac{1}{4} \cdot 0}_{\substack{\text{one child} \\ \text{and have} \\ \text{morethan} \\ \text{3children}}} + \frac{1}{2}\left[\underbrace{\left(\frac{1}{2}\right)^2}_{2/2} + \underbrace{\frac{1}{4} \cdot \frac{1}{4} \cdot 2}_{(1,3) \text{ or } (3,1)}\right] + \frac{1}{4}\left[3\underbrace{\left(\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{2}\right)}_{1}\right]$$



$= \frac{1}{8} + \frac{1}{16} + \frac{3}{128} = \frac{27}{128}$

Similarly, 2 / 2 denotes each child having two children; '0' indicated that the child can have a maximum

of 3 children

$2 \cdot \frac{1}{4} \cdot \frac{1}{4}$ denotes each child having 1 and 3 or 3 and 1 children $= \frac{16}{128} + \frac{8}{128} + \frac{3}{128} = \frac{27}{128}$

Q.58 Answer:

5

Solution:

$\mathbf{L}$: Expression reduces to $2 \operatorname{cosec} 8x \quad \Rightarrow \mathbf{L = 2}$

M: We have

$$\frac{4\left(\cos^3 10^\circ + \sin^3 20^\circ\right)}{\left(\cos 10^\circ + \sin 20^\circ\right)} = \frac{4\cos^3 10^\circ + 4\sin^3 20^\circ}{\cos 10^\circ + \sin 20^\circ} = \frac{\cos 30^\circ + 3\cos 10^\circ + 3\sin 20^\circ - \sin 60^\circ}{\cos 10^\circ + \sin 20^\circ}$$

$$= \frac{3\left(\cos 10^\circ + \sin 20^\circ\right)}{\left(\cos 10^\circ + \sin 20^\circ\right)} = 3 = \mathrm{M}$$

Hence $L + M = 2 + 3 = 5$

Q.59 Answer:

Has infinite number of solutions, real or non-real

Solution:

1st two columns of 1st determinant are same as 1st two rows of 2nd . Hence transpose the 2nd . Add the two determinants and use $C_1 \to C_1 + C_3 \Rightarrow D = 0$

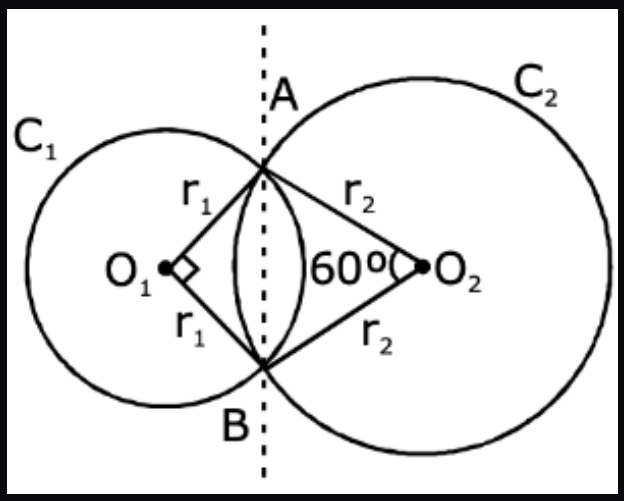
Q.60 Answer:

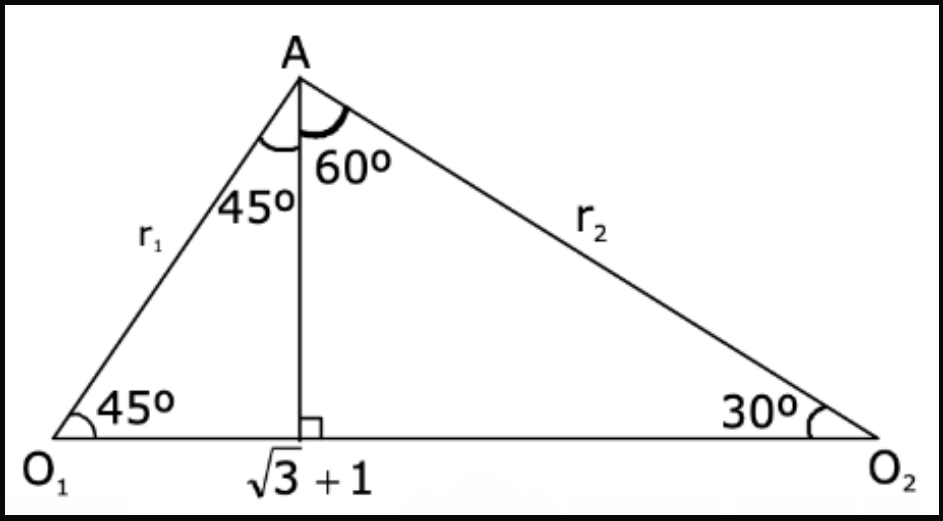
$\sqrt{2}$ and 2

Solution:

$\mathrm{O_1O_2} = \sqrt{3} + 1$

Sine rule in $\mathrm{AO_1O_2}$





$$\frac{\sqrt{3}+1}{\sin 105^\circ} = \frac{r_1}{\sin 30^\circ} = \frac{r_2}{\sin 45^\circ}$$

$$r_1 = \frac{\sqrt{3}+1}{\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)} \times \frac{1}{2} = \sqrt{2}$$

$$r_2 = 2$$

Q.61 Answer:

$[1, 16]$

Solution:

We have,

$f(x) = \sin^{-1}\left[\log_4\left(\frac{x}{4}\right)\right] + \sqrt{17x - x^2 - 16}$

$f(x)$ **is defined of**

$\log_4\left(\frac{x}{4}\right) \in [-1, 1]$ and $17x - x^2 - 16 \geq 0$

$\therefore 4^{-1} \leq \frac{x}{4} \leq 4$ and $x^2 - 17x + 16 \leq 0$

$\Rightarrow 1 \leq x \leq 16$ and $(x - 16)(x - 1) \leq 0$

$\Rightarrow 1 \leq x \leq 16$ and $1 \leq x \leq 16$

$\therefore$ Domain of $f(x)$ is $[1, 16]$.

Q.62 Answer:

5

Solution:

we have focus $(-1, -1)$ **and corresponding directrix** $x - y + 3 = 0$ $e = \frac{1}{2}$ **(given)**

Now, equation of axis of ellipse is

$x + y + k = 0$

As (1) passes through $(-1, -1)$**, so** $k = 2$

$\therefore$ **Equationof axis is**

$x + y + 2 = 0 \ldots\ldots\ldots (3)$

∴ on solving (1) and (3), we get foot of directrix

$\left(\frac{-5}{2}, \frac{1}{2}\right)$

Now, distance between focus and corresponding foot of directrix

$$= \left(\frac{a}{e} - ae\right) = \sqrt{\left(\frac{-5}{2}+1\right)^2 + \left(\frac{1}{2}+1\right)^2} \quad \left(\text{as } e = \frac{1}{2}\right)$$

$$\Rightarrow 2a - \frac{a}{2} = \frac{3}{\sqrt{2}} \Rightarrow a = \sqrt{2}$$

Also equation of mirror axis is

$x - y + \lambda = 0$

As distance of minor axis from focus is ae, so

$\frac{|-1+1+\lambda|}{\sqrt{2}} = \sqrt{2}\left(\frac{1}{2}\right) \Rightarrow |\lambda| = 1 \Rightarrow \lambda = \pm 1$

Q.63 Answer:

2

Solution:

lies $\alpha = a + ib, \gamma = c + id$

$\sqrt{a^2+b^2} + \sqrt{c^2+d^2} = p$

$f(z) = (4+i)z^2 + z(a+ib) + (c+id)$

$f(1) = (4+a+c) + i(1+b+d)$

hence $1 + b + d = 0$ **...(1)**

$f(i) = -(4+i) + i(a+ib) + (c+id)$

$-4 - b + c + i(a+d-1)$

hence $a + d = 1$ **...(2)**

\text { from (1) & (2) } a-b=2

hence there is no restriction of 'c' let $c = 0$

$|\alpha| + |\gamma| = \sqrt{a^2+b^2} + \sqrt{d^2} = \sqrt{4+2ab} + |d| \leq \sqrt{4+2ab} \geq \sqrt{2}$

with equality

$d = 0, a = 1, b = -1$

hence $p = \sqrt{2}$

Q.64 Answer:

0

Solution:

$$2\ln\left(\mathrm{f(x)} + \sqrt{\mathrm{f^2(x)}+1}\right)\mathrm{f'(x)} \leq \sqrt{\mathrm{f^2(x)}+1}$$

$$\int_\lambda^\mu \frac{2\ln\left(\mathrm{f(x)}+\sqrt{\mathrm{f^2(x)}+1}\right)\mathrm{f'(x)dx}}{\sqrt{\mathrm{f^2(x)}+1}} \leq \int_\lambda^\mu \mathrm{dx}; \qquad \left[\ln^2\left(\mathrm{f(x)} + \sqrt{\mathrm{f^2(x)}+1}\right)\right]_\lambda^\mu \leq (\mu - \lambda)$$

$$\mathrm{Lim_{x\to\mu^-}}\ln^2\left(\mathrm{f(x)} + \sqrt{\mathrm{f^2(x)}+1}\right) - \mathrm{Lim_{x\to\lambda^+}}\ln^2\left(\mathrm{f(x)} + \sqrt{\mathrm{f^2(x)}+1}\right) \leq \mu - \lambda;$$

$$\ln^2(3+\sqrt{10}) - \ln^2(\sqrt{10}-3) \leq \mu - \lambda; \quad 0 \leq \mu - \lambda \Rightarrow \text{ mm is value of } \mu - \lambda = 0$$

Q.65 Answer:

$e > \frac{2}{\sqrt{3}}$

Solution:

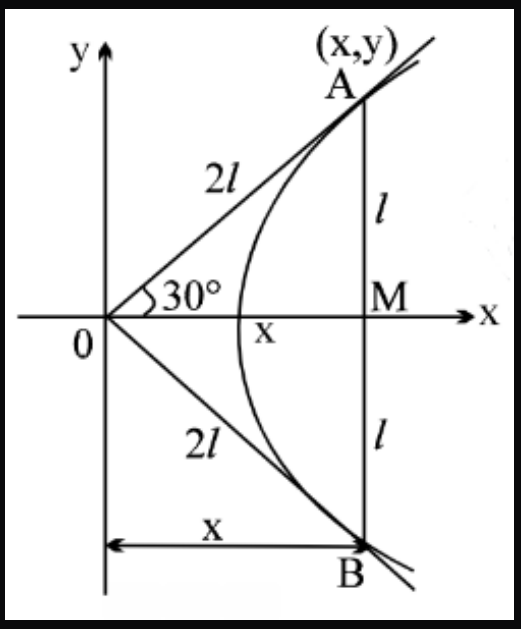
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \text{ where } y = l$$

$$\frac{x^2}{a^2} = 1 + \frac{l^2}{b^2}$$

$$\Rightarrow x^2 = \left(b^2 + R\right) \frac{a^2}{b^2} \quad \ldots(1)$$

$$\text{now, } x^2 + R = 4R$$

$$\Rightarrow x^2 = 3R \quad \ldots(2)$$



$$\text{from (1) and (2) } \frac{a^2\left(b^2 + l^2\right)}{b^2} = 3l^2 \Rightarrow a^2 b^2 + a^2 l^2 = 3 b^2 l^2$$

$$l^2\left(3 b^2 - a^2\right) = a^2 b^2$$

$$l^2 = \frac{a^2 b^2}{3 b^2 - a^2} > 0$$

$$\Rightarrow 3 b^2 - a^2 > 0 \Rightarrow \frac{b^2}{a^2} > \frac{1}{3}$$

$$1 + \frac{b^2}{a^2} > \frac{4}{3}$$

$$\Rightarrow e^2 > \frac{4}{3}$$

$$\Rightarrow e > \frac{2}{\sqrt{3}}$$

Q.66 Answer:

$a^2 + b^4$ is a perfect square

Solution:

$P^2 = I - 2P$

$P^3 = P - 2P^2 = P - 2(I - 2P)$

$= P - 2I + 4P$

$P^3 = 5P - 2I$

$P^4 = 5P^2 - 2 = 5(I - 2P) - 2P = 5I - 12P$

$P^2 + P^3 + P^4 = 4I - 9P$

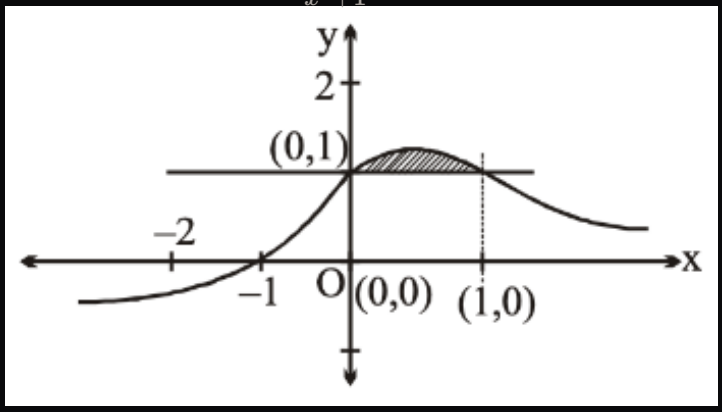
$\Rightarrow a^2 = 4 : b^2 = 9$

Q.67 Answer:

$\frac{1}{2}\ln 2 + \frac{\pi}{4} - 1$

Solution:

On solving $C : y = \frac{x+1}{x^2+1}$ and the line $L : y = 1$, we have



$\frac{x+1}{x^2+1} = 1 \Rightarrow x + 1 = x^2 + 1 \Rightarrow x(x-1) = 0 \Rightarrow x = 0, 1$

Thus, the points of intersection are at $x = 0$ and $x = 1$.

$$\text{Required area } = \int_0^1 \left(\frac{x+1}{x^2+1} - 1\right) dx = \frac{1}{2}\int_0^1 \frac{2x}{x^2+1} dx + \int_0^1 \frac{1}{x^2+1} dx - \int_0^1 dx$$

$$= \left[\frac{1}{2}\ln\left(x^2+1\right) + \left(\tan^{-1} x\right) - x\right]_0^1 = \frac{1}{2}\ln 2 + \frac{\pi}{4} - 1$$

Q.68 Answer:

$y^3 = -2x$

Solution:

$$\text{Here, } \quad \left(xy^3 - x^2\right) dy = \left(xy + y^4\right) dx$$

$$\Rightarrow \quad y^3(xdy - ydx) - x(xdy + ydx = 0)$$

$$\Rightarrow \quad x^2y^3 d\left(\frac{y}{x}\right) - xd(xy) = 0$$

dividing by x^3y^2, we get

$$\Rightarrow \quad \frac{y}{x} d\left(\frac{y}{x}\right) - \frac{1}{x^2y^2} \cdot d\left(\frac{x}{y}\right) = 0$$

$$\Rightarrow \quad \frac{1}{2} d\left(\frac{y}{x}\right)^2 + d\left(\frac{1}{xy}\right) = 0$$

Now integrating, we get

$\frac{1}{2}\left(\frac{y}{x}\right)^2 + \frac{1}{xy} = c$

It passes through $(4, -2)$

$\Rightarrow \quad \frac{1}{8} - \frac{1}{8} = c \Rightarrow C = 0$

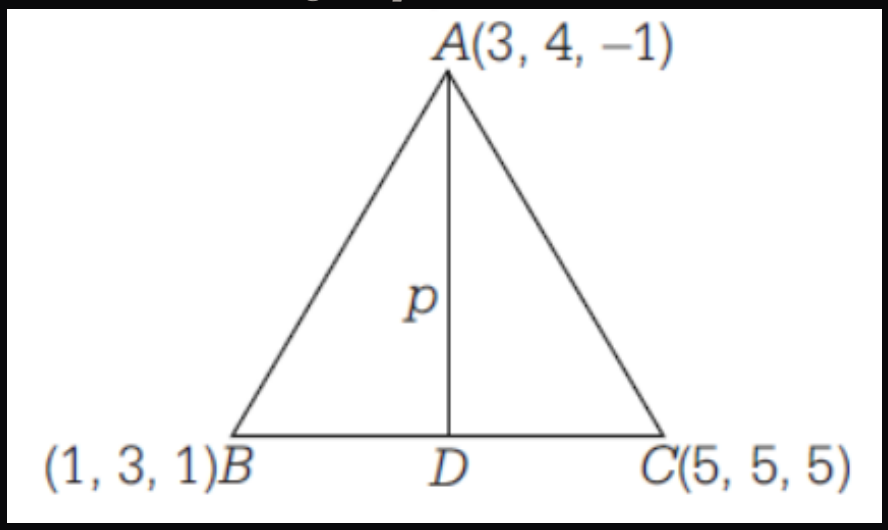
$\therefore y^3 = -2x$ is required curve.

Q.69 Answer:

$\frac{4}{3}\sqrt{5}$

Solution:

Let altitude of a triangle be p.



Now,

$$\mathrm{AB} = -2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$$

$$\mathrm{AC} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} + 6\hat{\mathbf{k}}$$

$$\mathrm{BC} = 4\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$$

and

We know that,

Area of $\triangle ABC = \frac{1}{2}|\mathrm{AB} \times \mathrm{AC}| = \frac{1}{2}|\mathrm{BC}| \times p$

Now, $\mathrm{AB} \times \mathrm{AC} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -2 & -1 & 2 \\ 2 & 1 & 6 \end{vmatrix} = -8\hat{\mathbf{i}} + 16\hat{\mathbf{j}} + 0\hat{\mathbf{k}}$

From Eq. (i),

$$\frac{1}{2}\sqrt{(-8)^2 + (16)^2 + (0)^2} = \frac{1}{2}\sqrt{4^2 + 2^2 + 4^2} \times p$$

$$\Rightarrow \quad p = \frac{\sqrt{320}}{6} = \frac{8\sqrt{5}}{6} = \frac{4\sqrt{5}}{3}$$

Q.70 Answer:

1

Solution:

Vertex is at $x = b$. Also $f(x)$ is decreasing in $(-\infty, b)$ and increasing in (b, ∞). $x = b$ is the point of minima.

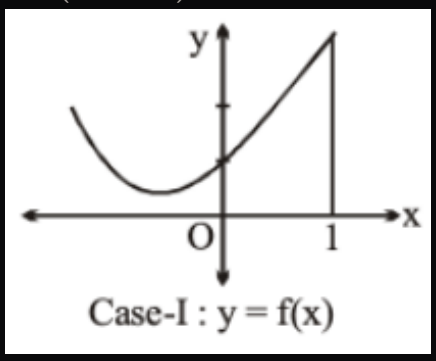
We have, $f(x) = x^2 - 2bx + 1$

Case I:

Let $b \leq 0$

So, $f(1) - f(0) = 4$

$\Rightarrow (2 - 2b) - 1 = 4$



$\Rightarrow 1 - 2b = 4 \Rightarrow b = \frac{-3}{2}$ **(which is** < 0**)**

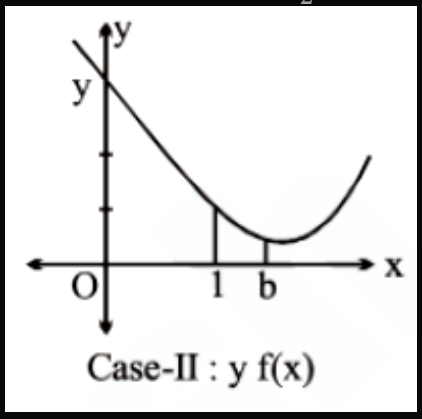
Case II:

Let $b \geq 1$

So, $f(0) - f(1) = 4$

$\Rightarrow 1 - (2 - 2b) = 4$

$\Rightarrow 2b = 5 \Rightarrow b = \frac{5}{2}$ **(which is** > 1**)**



Case III:

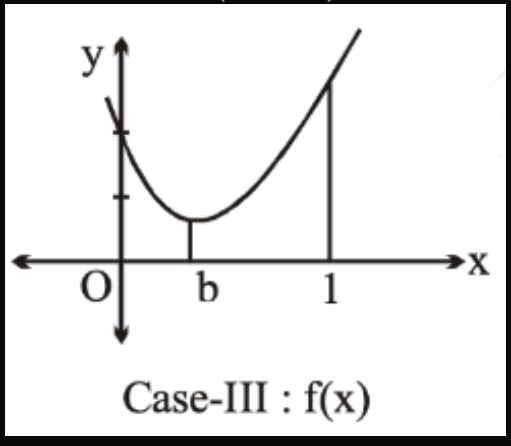
If $0 < b < 1$

Clearly, max. $[f(0), f(1)] - f(b) = 4$

Here, $f(1) > f(0)$

Hence, $f(1) - f(b) = 4$

$\Rightarrow 2 - 2b - \left(1 - b^2\right) = 4$



$\Rightarrow b^2 - 2b + 1 = 4$

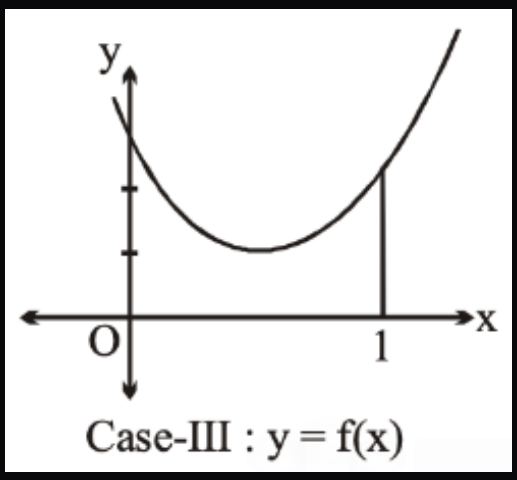
$\Rightarrow b - 1 = 2 \text{ or } -2$

$\Rightarrow b = 3 \text{ or } b = -1 \text{ (Rejected)}$

Here $f(0) > f(1)$

So, $f(0) - f(b) = 4$

$\Rightarrow 1 - (1 - b^2) = 4$



$\Rightarrow b = 2$ **or -2 (Rejected)**

Hence $b \in \left\{\frac{-3}{2}, \frac{5}{2}\right\}$

Hence, sum $= \frac{-3}{2} + \frac{5}{2} = 1$

Q.71 Answer:

—

Solution:

For continuity at $x = 1$ **we leave** $f(1^-) = 1$ **and** $f(1^+) = a + b + c$

$\therefore a + b + c = 1 \quad \ldots(1)$

$$f'(x) = \begin{bmatrix} 3x^2 & \text{if } x < 1 \\ 2ax + b & \text{if } x \geq 1 \end{bmatrix}$$

for continuity of $f'(x)$ **at** $x = 1$

$f'(1^-) = 3; f'(1^+) = 2a + b$

hence $2a + b = 3$ **...(2)**

$$f''(x) = \begin{bmatrix} 6x & \text{if } x < 1 \\ 2a & \text{if } x \geq 1 \end{bmatrix}$$

$f''(1^-) = 6; f''(1^+) = 2a$

$\therefore 2a = 6$

$\Rightarrow a = 3$

from (2), $b = -3$

from (2), $c = 1$

Hence $a = 3, b = -3; c = 1$

$\therefore \quad \sum a^3 = 1$

Q.72 Answer:

—

Solution:

We know that $(a-1)^n = {}^nC_0a^n - {}^nC_1a^{n-1} + {}^nC_2a^{n-2} - \ldots\ldots\ldots + (-1)^{n-1}{}^nC_{n-1}a + (-1)^nC_n$

$\therefore \quad \frac{(a-1)^n}{a} = {}^nC_0a^{n-1} - {}^nC_1a^{n-2} + {}^nC_2a^{n-3} - \ldots\ldots\ldots\ldots + (-1)^{n-1}{}^nC_{n-1} + \frac{(-1)^n}{a}{}^nC_n$

Hence $f(n) = \frac{(a-1)^n-(-1)^n}{a}$

Now, $f(2007) + f(2008) = \frac{(a-1)^{2007}+1}{a} + \frac{(a-1)^{2008}-1}{a} = \frac{(a-1)^{2007}(1+a-1)}{a} = (a-1)^{2007}$

$= \left(3^{\frac{1}{223}}\right)^{2007} = 3^9 = 3^2 \cdot 3^7 = 9(2187)$

Hence k = 2187

Q.73 Answer:

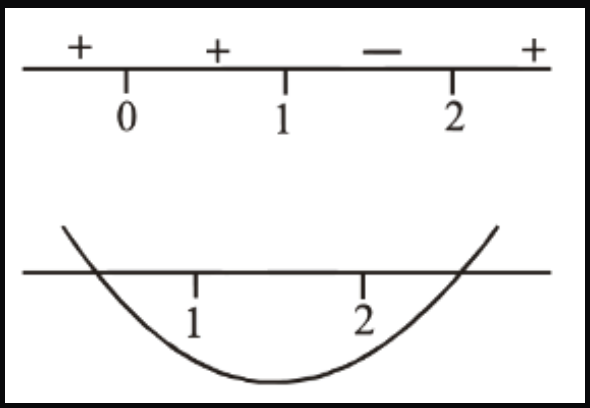
—

Solution:

$x(3x+2)^2 + 2 = (a+12+9x)x^2 - bx + c$

$\Rightarrow 4x + 2 = ax^2 - bx + c$

which is quadratic in x but it satisfies by α, β and γ so it is an identity.



$\therefore a=0, b=-4, c=2$

$x^2(4x-4)(x-2)<0$

$\Rightarrow x^2(x-1)(x-2)<0$

$x \in (1,2)$

$3x^2+px+p^2+6p<0$ satisfy by every $x \in (1,2)$

$f(1) \leq 0$ and $f(2) \leq 0$

$p^2+7p+3 \leq 0$

and $p^2+8p+12 \leq 0$

$p \in \left[\frac{-7-\sqrt{37}}{2}, \frac{-7+\sqrt{37}}{2}\right]$

$(p+2)(p+6) \leq 0$

$\Rightarrow p \in [-6,-2]$

$p \in [-6.5\ldots,-0\ldots]-6,-5,-4,-3,-2$

$\therefore$ **Number of integers are 5**

Q.74 Answer:

—

Solution:

Dividing the given d.e by $y^2\cos x$, $-\frac{1}{y^2}\frac{dy}{dx}+\sec x \cdot \frac{1}{y}=1-\sin x$

$\frac{d}{dx}\left(\frac{1}{y}\right)+\sec x \cdot \left(\frac{1}{y}\right)=1-\sin x$

Linear d.e with I.F. $=\exp \int \sec x dx=\sec x+\tan x$

Solution, $\frac{\sec x+\tan x}{y}=\int \frac{(1-\sin x)(1+\sin x)dx}{\cos x}=\int \cos x dx=\sin x+c$

$x=0, y=1 \Rightarrow c=1$

$\therefore \quad y=\frac{\sec x+\tan x}{\sin x+1}=\sec x, y\left(\frac{\pi}{3}\right)=2.$

Q.75 Answer:

—

Solution:

We have $\operatorname{cosec}^{-1}\operatorname{cosec} 2>x^2-3x \Rightarrow x^2-3x-(\pi-2)<0 \Rightarrow \frac{3-\sqrt{1+4\pi}}{2}<x<\frac{3+\sqrt{1+4\pi}}{2}$

.

$\therefore$ **Positive integral solution of above inequality are** $1,2,3.$

Now $2010=2\times3\times5\times67$**, total number of divisors of** $2010=16.$

So A contains 3 elements and B contains 16 elements. Number of mapping are as follows :

$f(1) < f(2) < f(3) \Rightarrow {}^{16}C_3 = 560$

$f(1) = f(2) < f(3) \Rightarrow {}^{16}C_2 = 120$

$f(1) < f(2) = f(3) \Rightarrow {}^{16}C_2 = 120$

$f(1) = f(2) = f(3) \Rightarrow {}^{16}C_1 = 16$

Hence total mappings $= 816$